Pulse Modulation

Sampling of Analog Signals
Quantization of Analog Signals
Pulse Code Modulation
Differential Pulse Code Modulation and Delta Modulation





The digital transmission of analog signals requires the analog signal to be digitized.

1. Sampling: Discretization on time axis
2. Quantization: Discretization of amplitude values
3. Encoding: Represents the finite amplitude values in binary code

Learning Guide

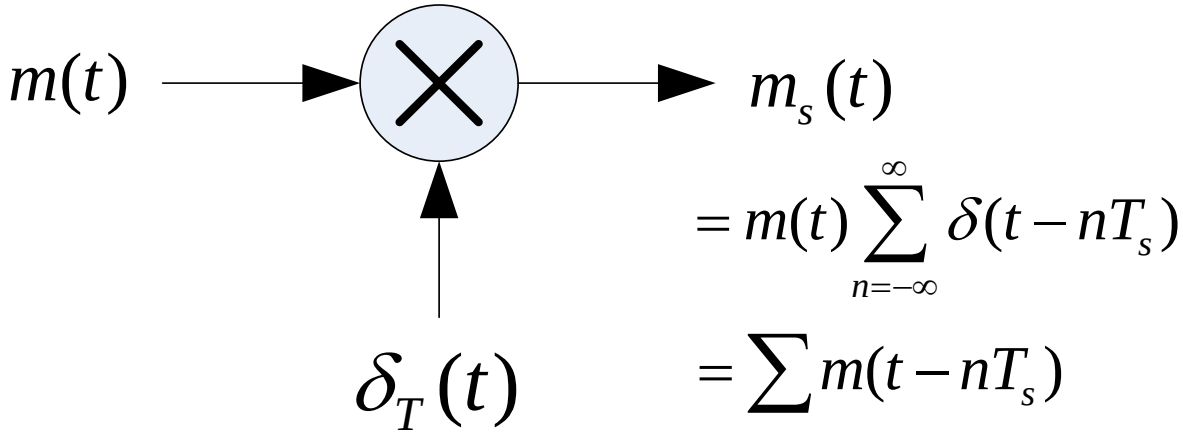
Sampling Theorem
Pulse Modulation

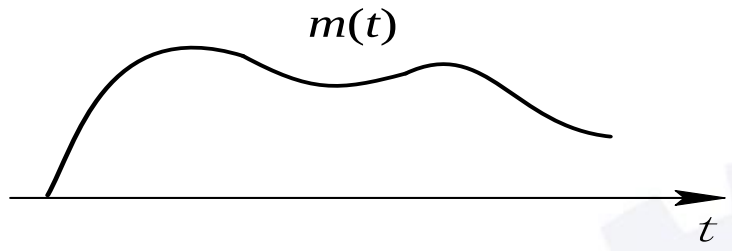
4.1 Sampling of Analog Signals

4.1.1 Sampling Theorem

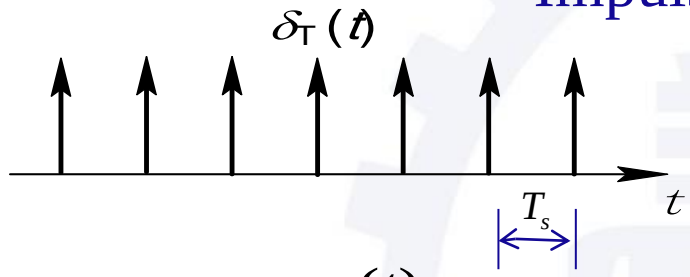
Sampling

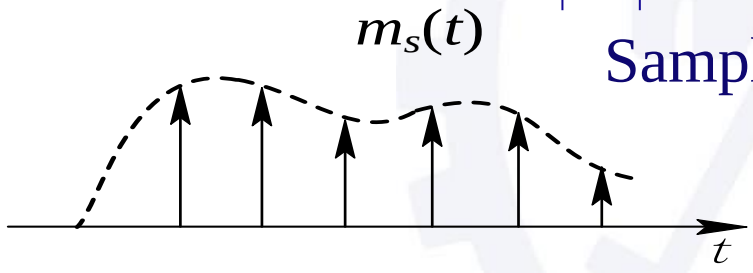
- Sampling: Convert time-continuous analog signals into time-discrete sampled values





Impulse sequence





Sampled value sequence

- Sampling interval T_s

Can the original signal $m(t)$ be completely restored by $m_s(t)$?

Low Pass Sampling Theorem

A low pass signal $m(t)$ with a frequency band limited to (0, f_m) can be completely restored from the sampled signal if the sampling frequency meets the following condition

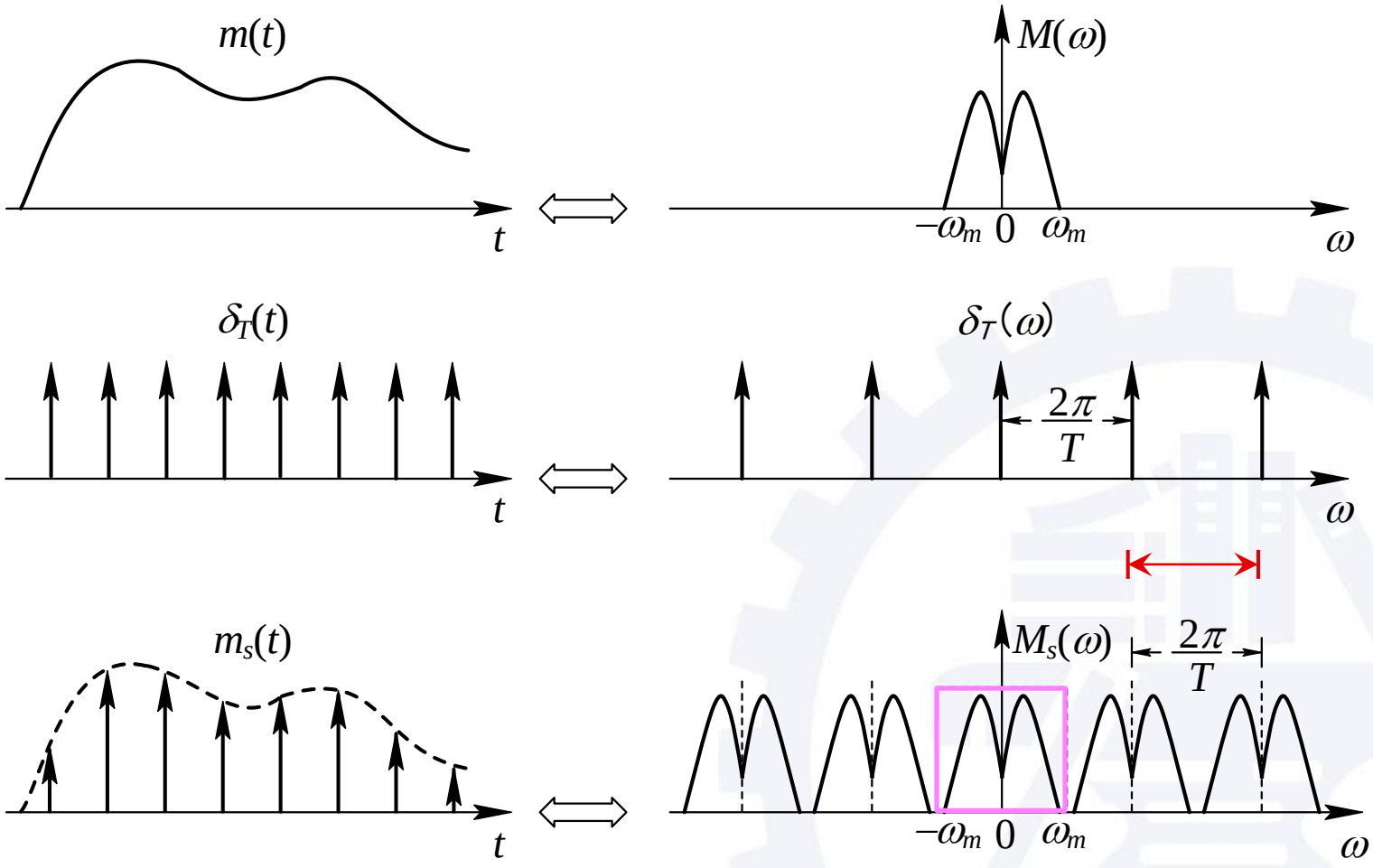
$$f_s \geq 2 f_m$$

Sampling Frequency: $f_s = \frac{1}{T_s}$

Sampling Interval (period): T_s

Nyquist Rate: $f_s = 2f_m$

Nyquist Interval: $T_s = 1/(2f_m)$



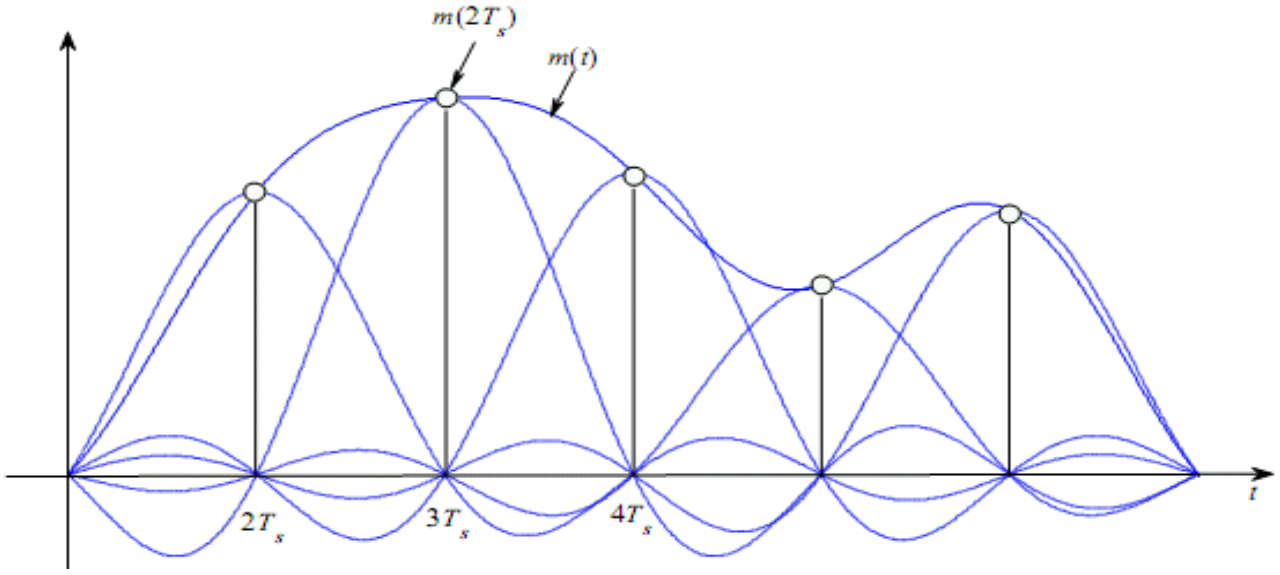
Restore continuous signal: Low pass filter

Restore Original Signal

Frequency domain: Filter out the baseband spectrum with a low pass filter

$$M(\omega) = T_s[M_s(\omega)H(\omega)]$$

$$H(\omega) = \text{rect}\left(\frac{\omega}{2\omega_m}\right)$$



Time domain: interpolating between sampled values with Sa(x)

$$m(t) = T_s[m_s(t) * h(t)]$$

$$= T_s \times [m(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s)] * \frac{\omega_m}{\pi}Sa(\omega_m t)$$

$$= \frac{1}{f_s}[\sum_{n=-\infty}^{\infty} m(nT_s)\delta(t-nT_s)] * \frac{\omega_m}{\pi}Sa(\omega_m t)$$

$$= \sum_{n=-\infty}^{\infty} m(nT_s)Sa[\omega_m(t-nT_s)]$$

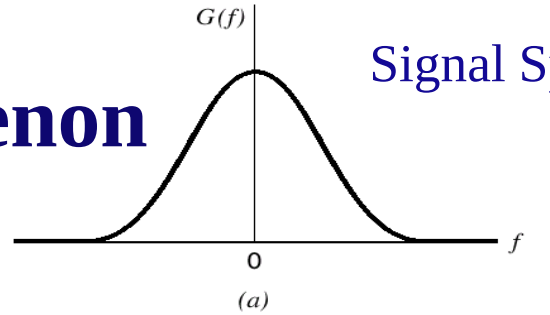
$m(nT_s)$: Sampled values of $m(t)$ at time $t = nT_s$,$n = 0, \pm 1, \pm 2, \ldots$

Sampling Distortion: Aliasing Phenomenon

- Aliasing Phenomenon
  - Sampling rate is too low
  - Message signals are not strictly band-limited
- Solutions:
  - Select a sampling rate slightly greater than the Nyquist rate

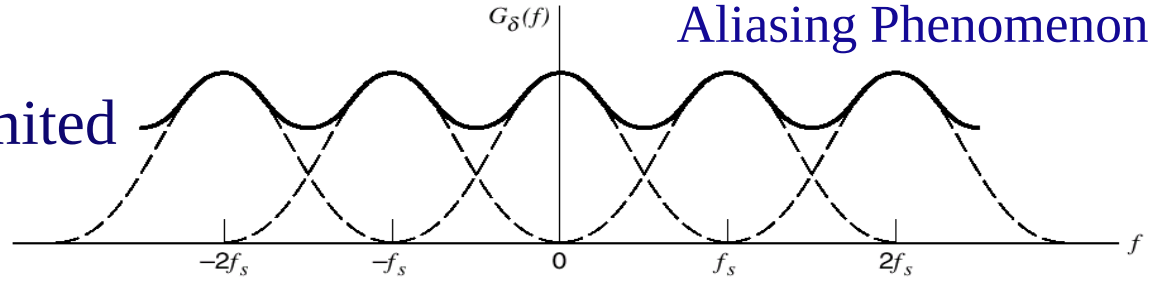
$$f_s > 2f_m$$

- Filter before sampling to prevent aliasing (LPF)
  - Limit the signal frequency not exceed f_m

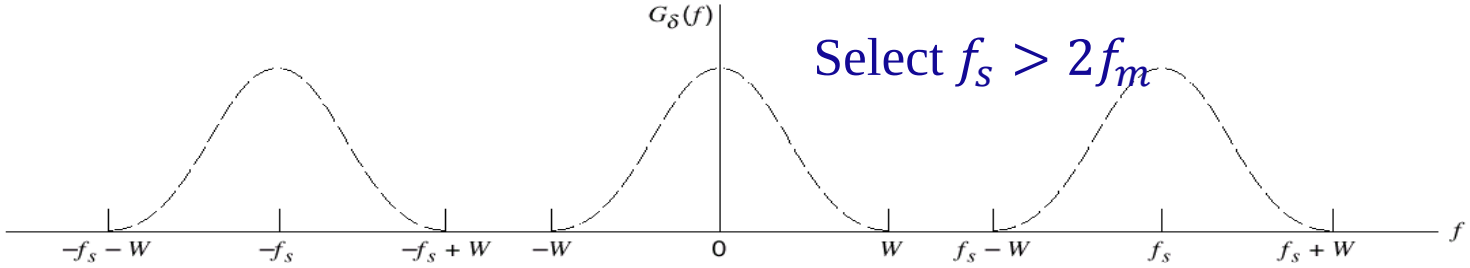


Signal Spectrum

(a)

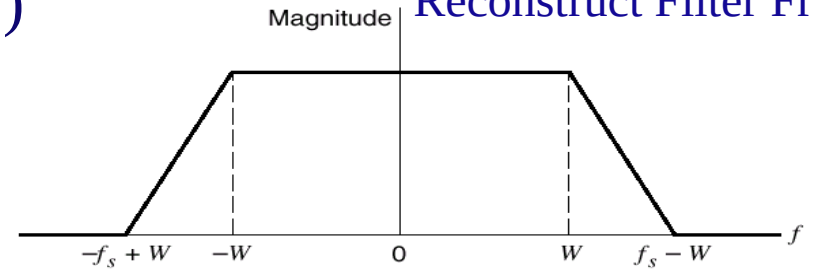


Aliasing Phenomenon



Select $f_s > 2f_m$

(b)

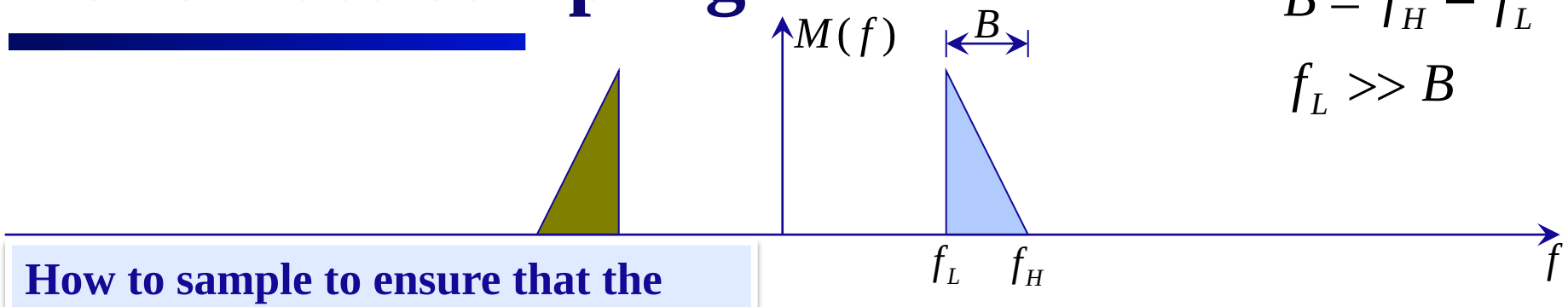


Reconstruct Filter Frequency Response

Band-Pass Sampling

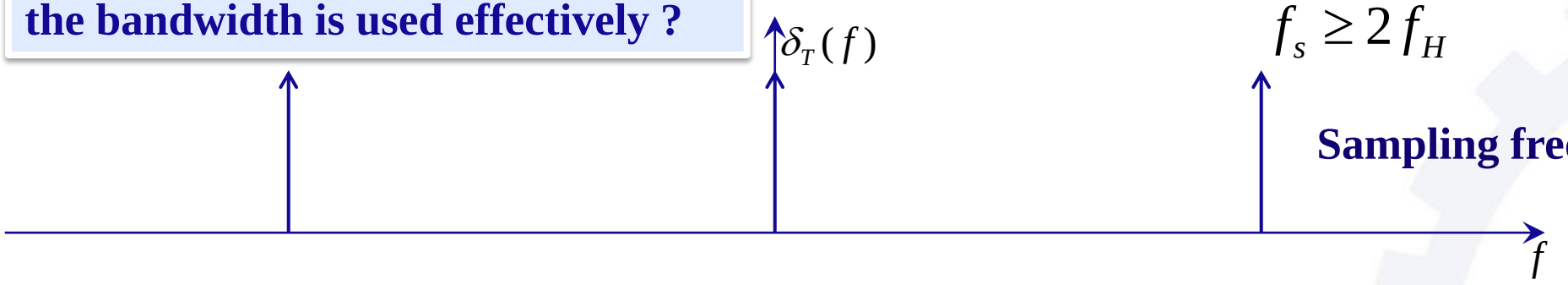
$$B = f_H - f_L$$

$$f_L >> B$$



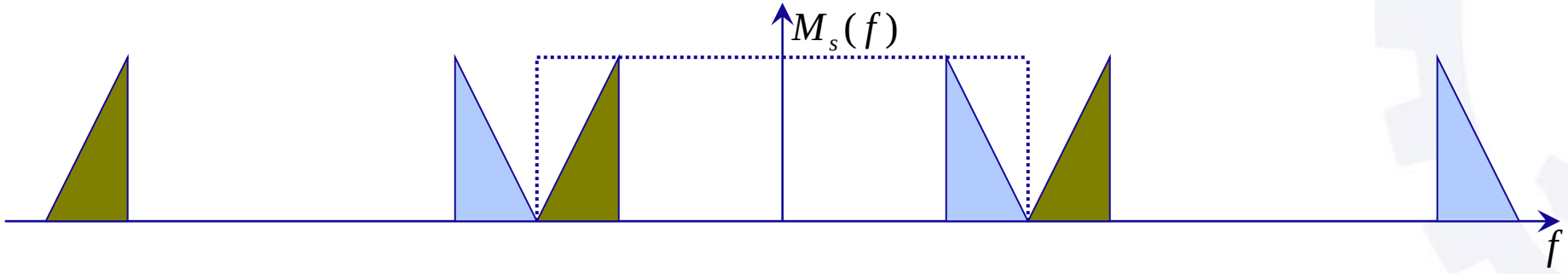
How to sample to ensure that the signal spectrum is not aliased and the bandwidth is used effectively ?

$$f_s \geq 2 f_H$$



Sampling frequency is too high

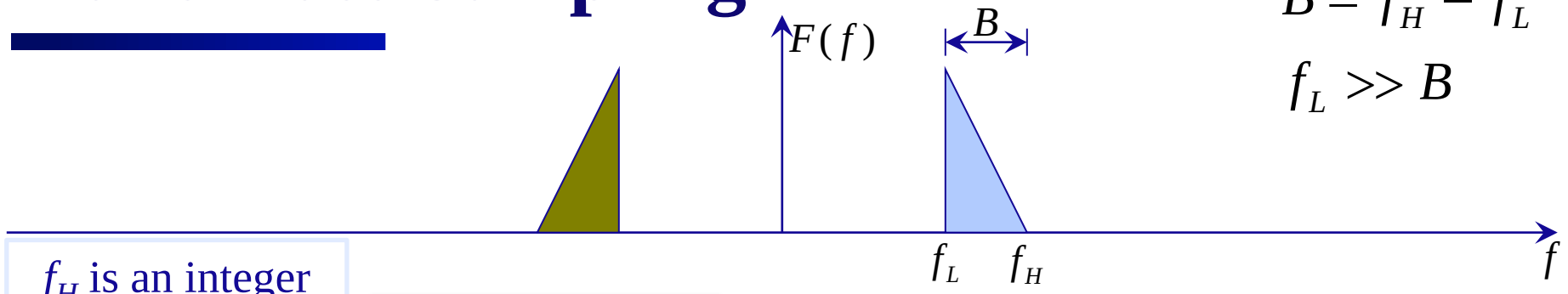
Result?



Band-Pass Sampling

$$B = f_H - f_L$$

$$f_L >> B$$

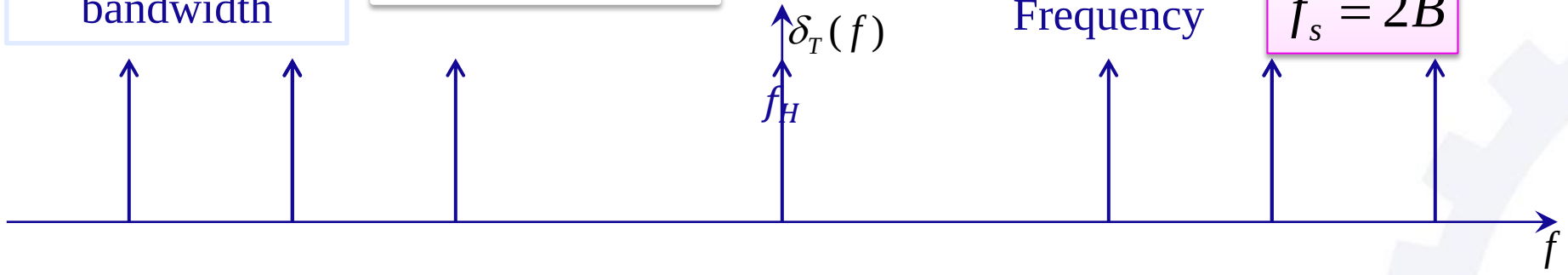


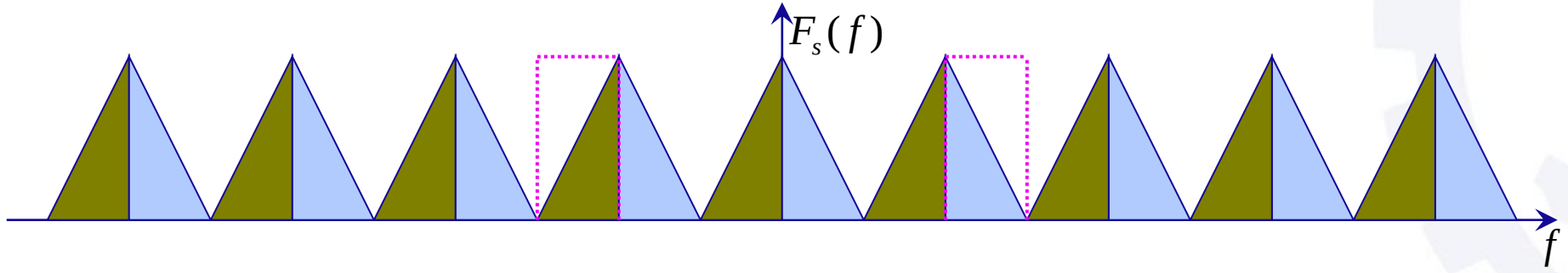
f_H is an integer multiple of the bandwidth

$$f_H = NB$$

Sampling Frequency

$$f_s = 2B$$





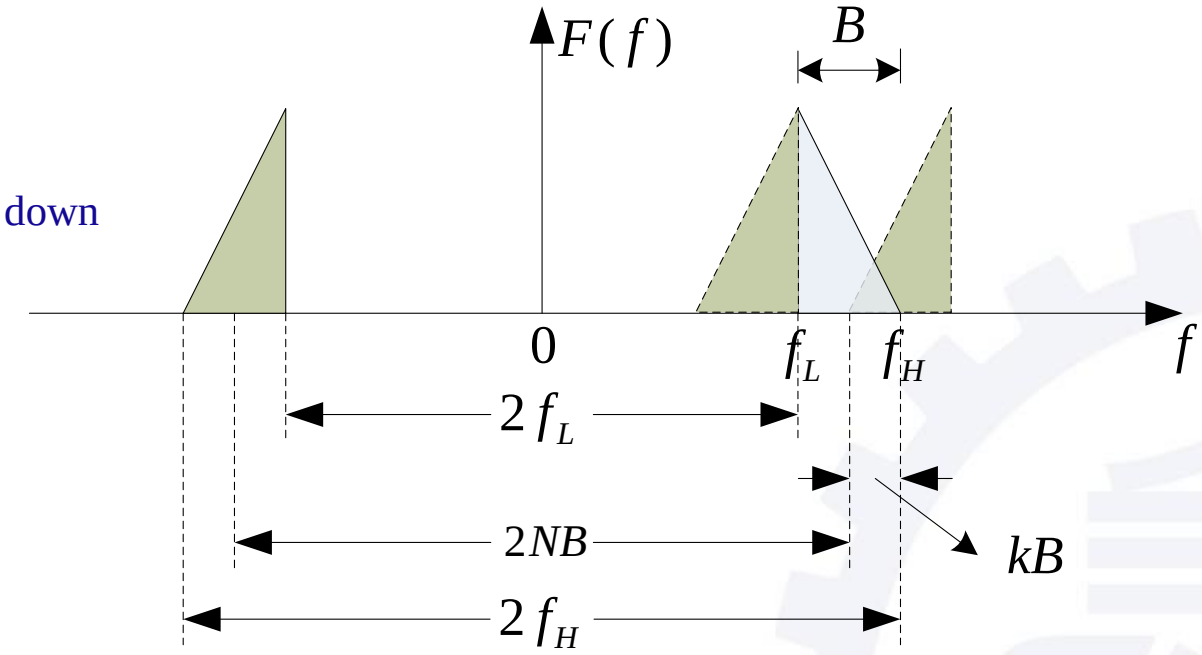
Band-Pass Sampling Theorem

f_H is a non-integer multiple of bandwidth

$$f_H = (N + k)B$$

$$N = \left\lfloor \frac{f_H}{B} \right\rfloor \quad \text{Round down}$$

$$0 \le k < 1$$



Band-pass signal distortion-free sampling rate

$$f_s \ge \frac{2f_H}{N} = 2B\left(1 + \frac{k}{N}\right)$$

Restore continuous signal: Band-pass filter

Sampling Theorem

Nyquist rate is 2 times of the highest frequency of low-pass signal

Low-pass signal distortion-free sampling rate is not lower than Nyquist rate

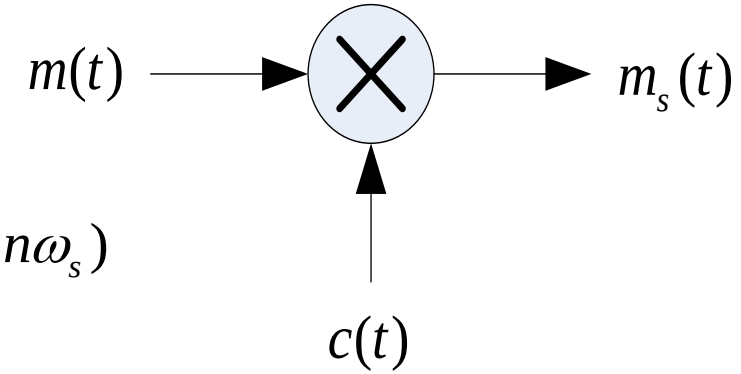
Band-pass signal distortion-free sampling rate is slightly higher than 2 times of signal bandwidth

4.1.2 Pulse Modulation

Natural Sampling

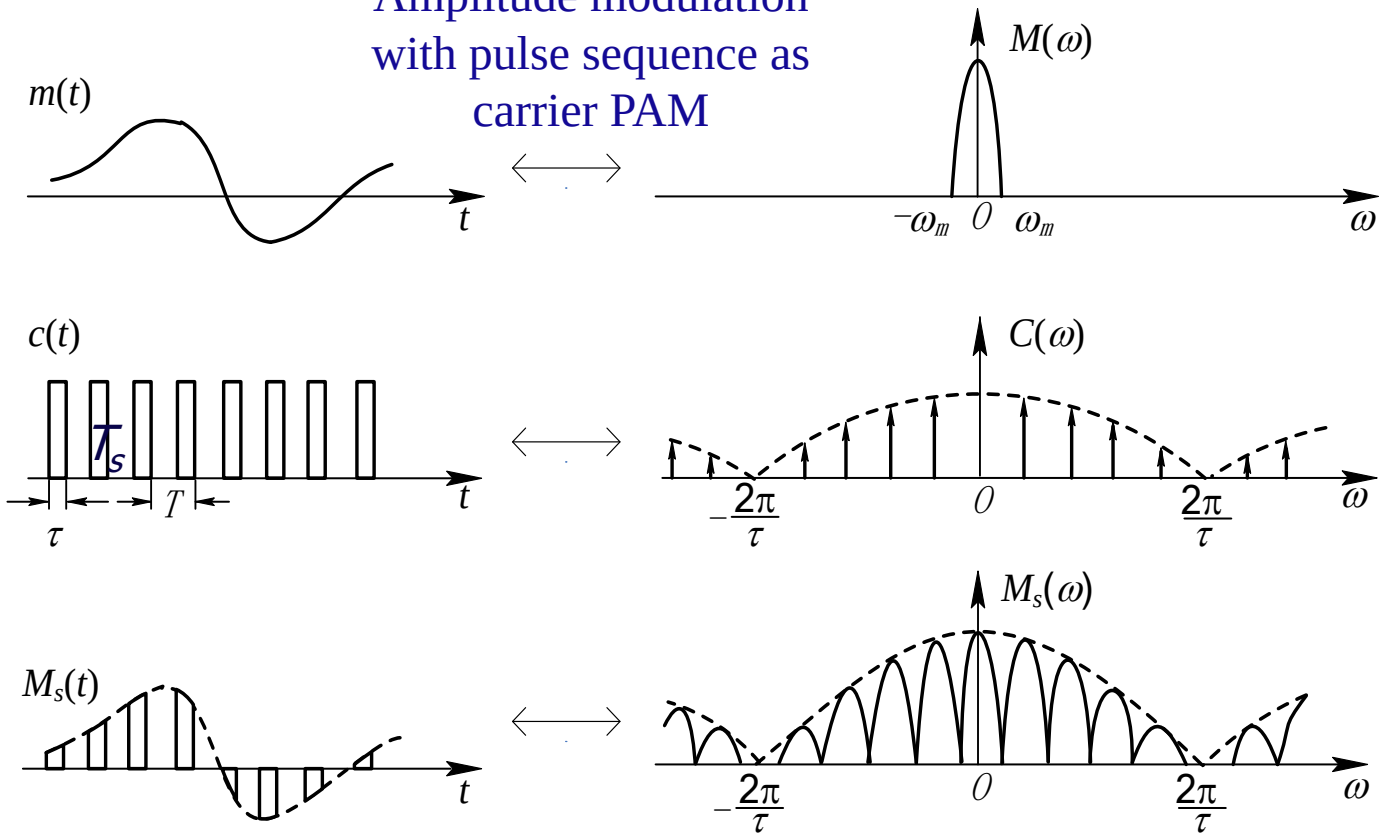
$$M_s(\omega) = \frac{1}{2\pi}[M(\omega) * C(\omega)]$$

$$= \tau \cdot f_s \cdot \sum Sa(\frac{n\tau\omega_s}{2}) M(\omega - n\omega_s)$$



- Actual Sampling
  - Sampling with narrow pulse sequence

Amplitude modulation with pulse sequence as carrier PAM



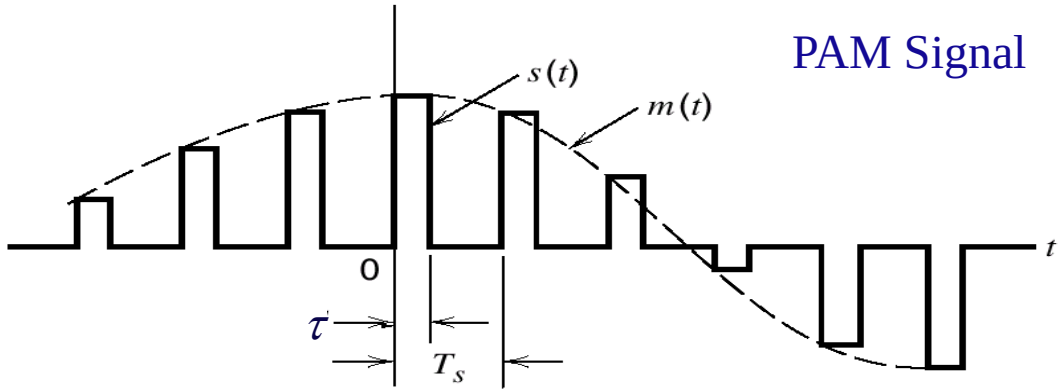
Flat-top Sampling

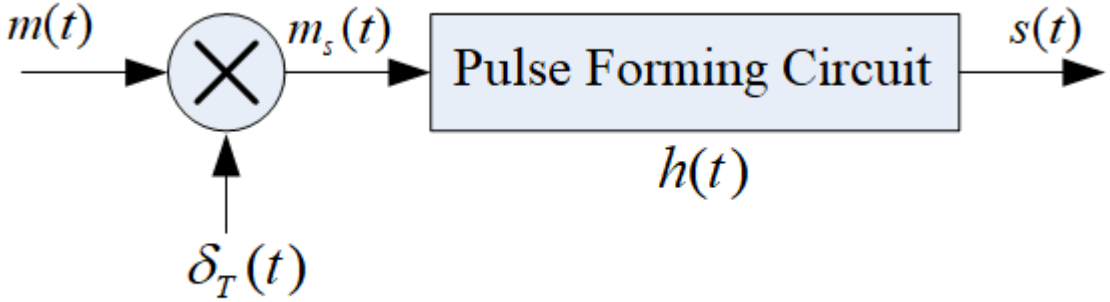
- Sample Hold
  - Instantaneous sample the signal $m(t)$ every T_s seconds
  - Extend the duration of each sampled value to τ

- Pulse Carrier: Duty Pulse Sequence
  - Amplitude
  - Duration
  - Position

- Pulse-Amplitude Modulation, PAM
  - Pulse amplitude varies in proportion to the sampled values of continuous signal

PAM Signal





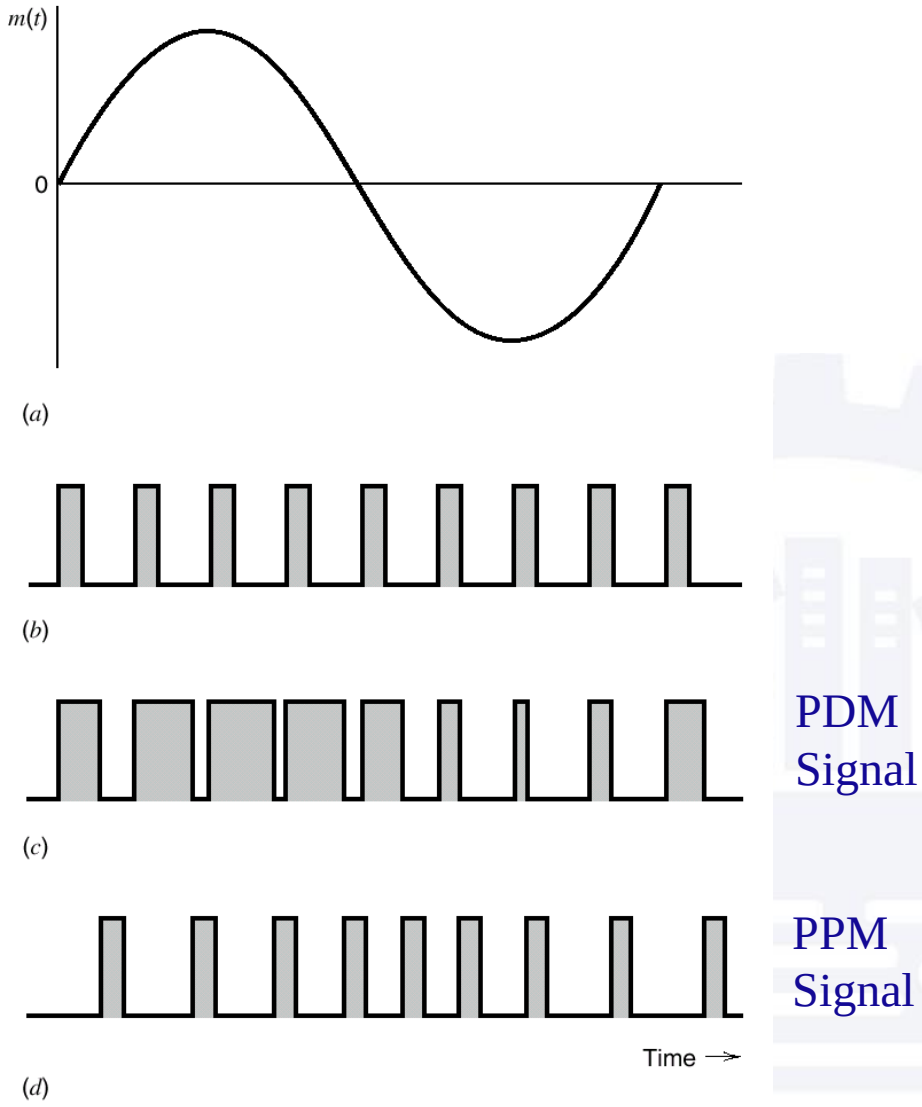
$$h(t) = \text{rect}\left(\frac{t - \tau/2}{\tau}\right)$$

$$S(\omega) = M_s(\omega) \cdot H(\omega)$$

$$= \tau \cdot f_s \cdot \sum M(\omega - n\omega_s) \cdot Sa\left(\frac{\omega\tau}{2}\right) \cdot e^{-j\frac{\omega\tau}{2}}$$

Analog Pulse Modulation

- PAM (Pulse-Amplitude Modulation)
  - Pulse amplitude varies in proportion to the sampled values of continuous signal
- PDM (Pulse-Duration Modulation)
  - Pulse duration (pulse width) varies with signal sampled values
  - PDM wastes power
- PPM (Pulse-Position Modulation)
  - Pulse position, relatives to the occurrence time of unmodulated pulse, varies with signal sampled values
  - Efficiency is better than PDM
  - Similar to analog phase modulation, threshold effect occurs



Digital Pulse Modulation

- Analog Pulse Modulation
  - The number of messages is infinite
  - Discrete on the time axis
- Digital Pulse Modulation
  - Discrete on the time axis
  - The number of messages is finite

Sampling

Quantization

Encoding

L Signal Amplitudes - Quantization Levels

L different binary bit combinations represent *L* different signals

Summary of Pulse Modulation

Pulse amplitude, duration and position can carry information

Digital pulse modulation has a finite number of messages

Uniform Quantization

Optimized Quantizer

4.2 Quantization of Analog Signals



4.2.1 Uniform Quantization

Quantization Process

- Determine the dynamic range $2A$ of signal $m(t)$

$$A = |m(t)|_{\max}, \quad -A \le m(t) \le A$$

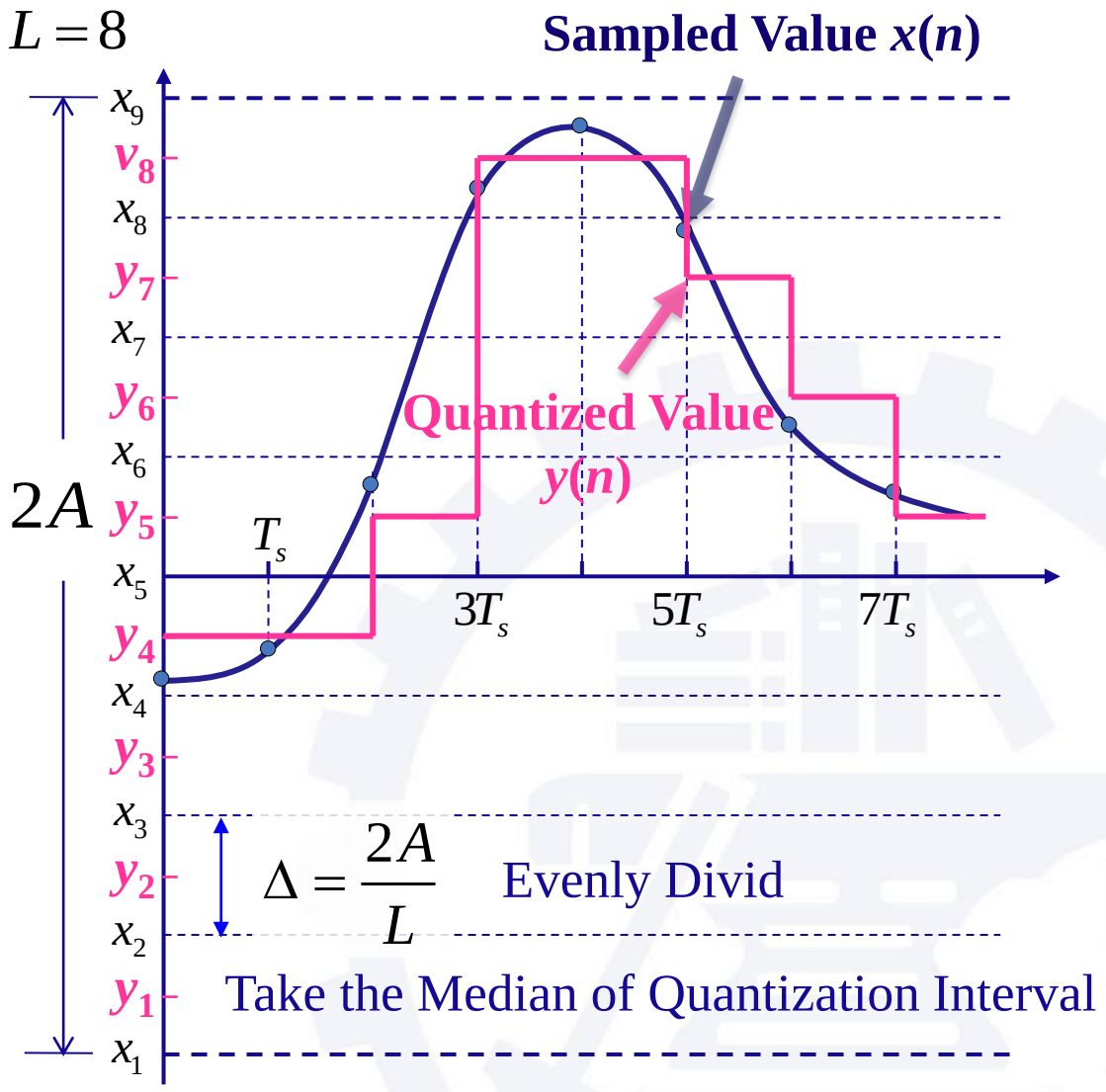
- Divid into L quantization intervals

$$I_i : \{x_i < x \le x_{i+1}\}, \quad i = 1, 2, ..., L$$

  - x_i: Layer Level (Interval Boundary)
  - $\Delta_i = x_i - x_{i-1}$: Quantization Interval

- Determine quantization / reconstruction level y_i
  - Represents all levels in the quantization interval I_i

- Quantization: The signal sampled values falling into the interval I_i are mapped to the quantization level y_i



Quantizer Parameters

- Determine the dynamic range $2A$ of signal $m(t)$

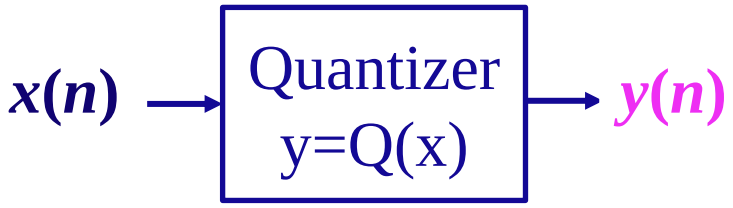
$$A = |m(t)|_{\max}, \quad -A \le m(t) \le A$$

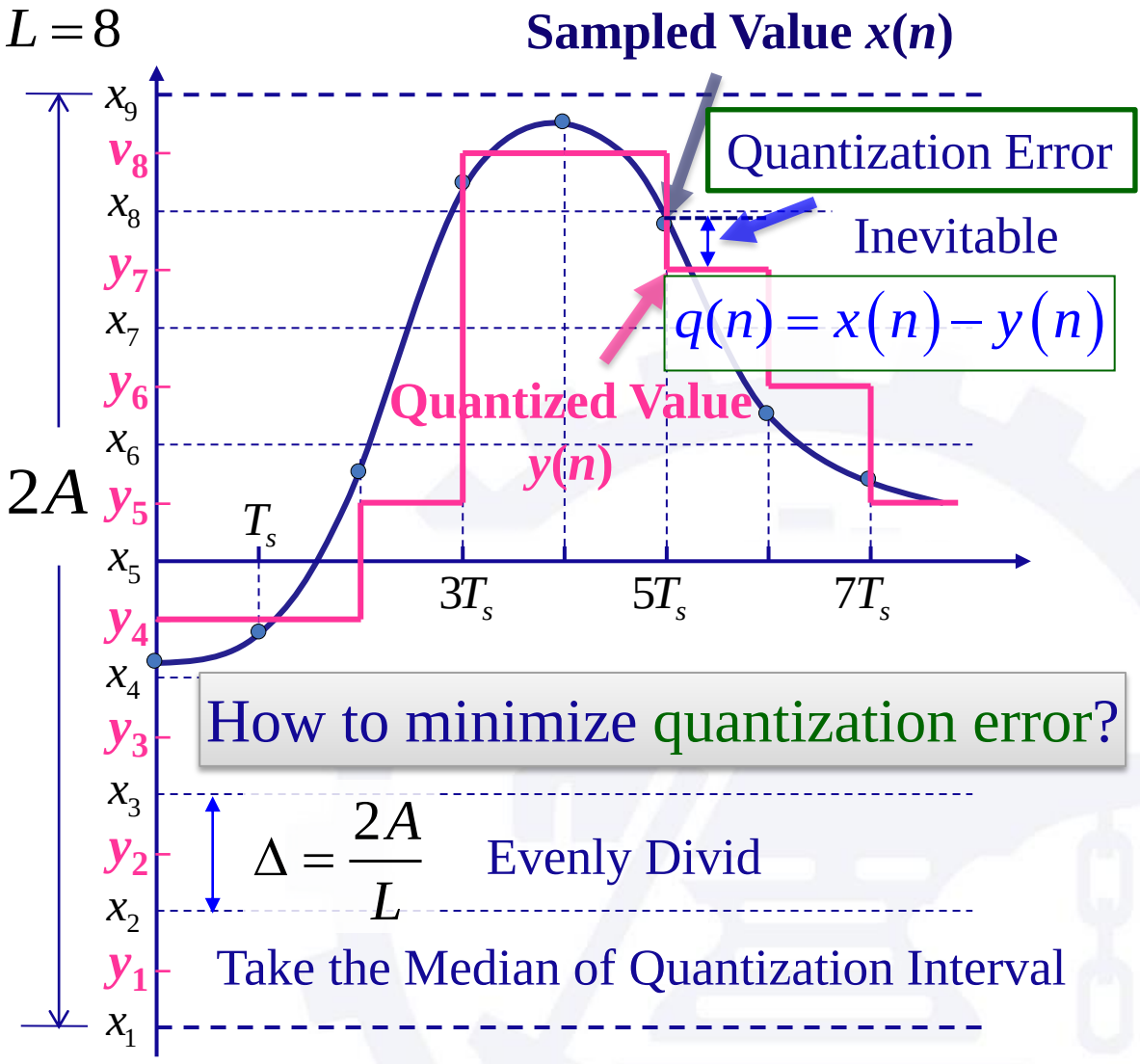
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- Determine quantization / reconstruction level y_i
  - Represents all levels in the quantization interval I_i





Quantified noise average power

$$N_q = E[q^2] = \int_{-A}^{A} \left(x - Q(x)\right)^2 p(x)dx$$

$$= \sum_{i=1}^{L} \int_{x_i}^{x_{i+1}} \left[x - y_i\right]^2 p(x)dx$$

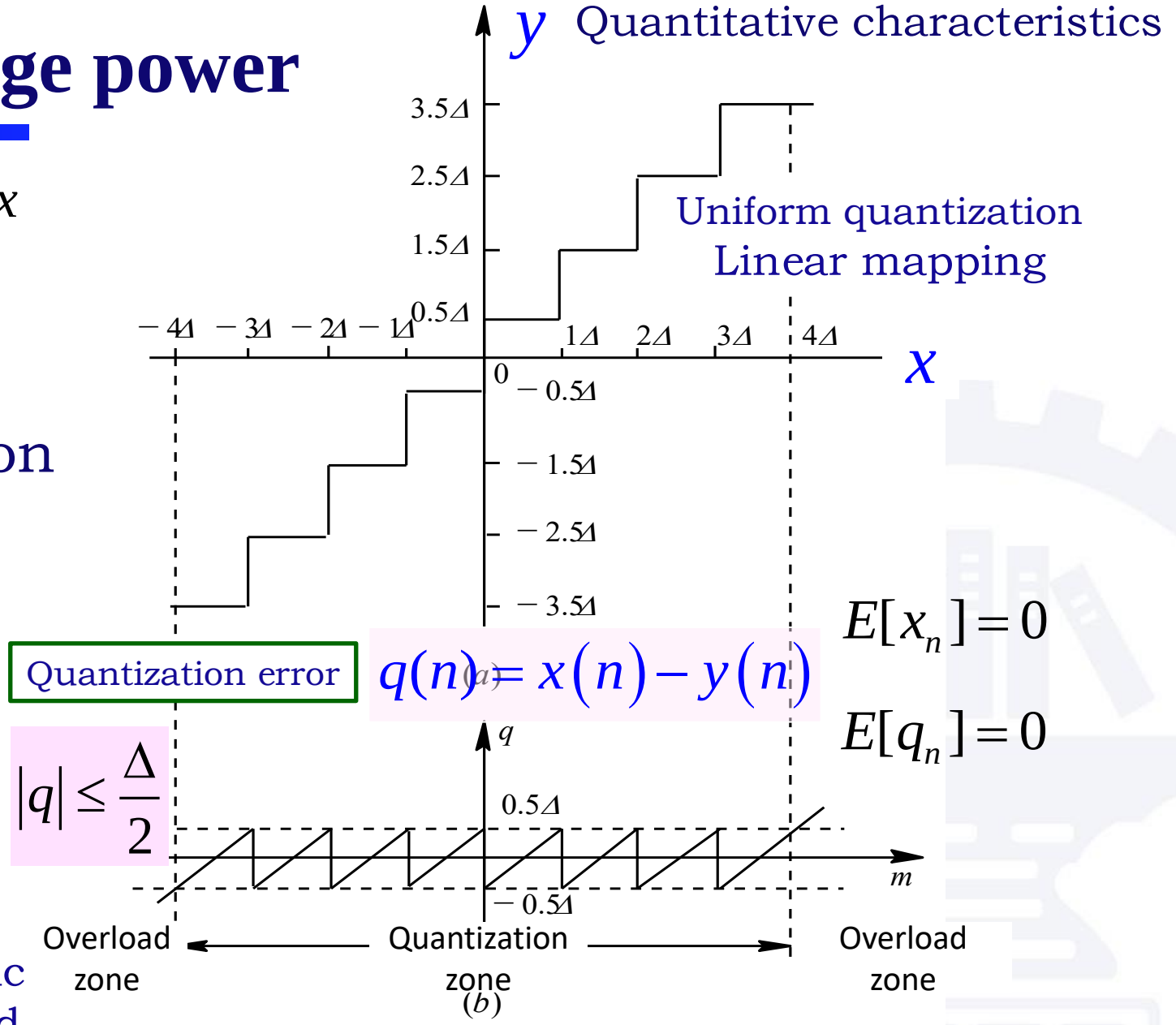
- Uniform signal distribution

$$N_q = \frac{1}{2A} \sum_{i=1}^{L} \int_{x_i}^{x_{i+1}} \left[x - y_i\right]^2 dx$$

- Uniform quantization

$$N_q = \frac{L}{2A} \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} q^2 dq = \frac{\Delta^2}{12}$$

The noise that exceeds the dynamic range [-A,A] is called overload noise



Quantified signal-to-noise ratio

$$\Delta = \frac{2A}{L} \qquad L = 2^k$$

- Quantified signal-to-noise ratio

$$\frac{S}{N_q} = \frac{\sigma_M^2}{\Delta^2/12} = \frac{3\times 2^{2k}}{k_{cr}^2}$$

  - Input signal power $S = \sigma_M^2$ Uniform quantization
  - Quantified noise power $N_q = \frac{L}{2A}\int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} q^2 dq = \frac{\Delta^2}{12}$

$$\frac{S}{N_{q\,\mathrm{dB}}} = 6.02k + 4.77 - 20\log_{10} k_{cr}$$

If the code is increased by 1 bit, the quantified SNR is increased by 6dB.

Crest factor $k_{\mathrm{cr}} = \frac{A}{\sigma_M}$

Sinusoidal signal $k_{\mathrm{cr}} = \sqrt{2}$

Triangle signal $k_{\mathrm{cr}} = \sqrt{3}$

Sinusoidal signal $\frac{S}{N_q} = 1.5\times 2^{2k} \approx 1.8 + 6k \ (\mathrm{dB})$

Summary of Uniform Quantization

The greater the number of stages divided by the quantization interval is and the smaller the interval is, the smaller the quantization noise is but the greater number of effective coding bits is.

If the code is increased by 1 bit, the quantified SNR is increased by 6dB.

4.2.2 Optimized Quantizer

Optimized Quantizer

Goal: Minimize quantization error

$$N_q = \sum_{i=1}^{L} \int_{x_i}^{x_{i+1}} [x - y_i]^2 \, p(x) dx$$

- **The actual signal is not uniformly distributed**
  - Voice: Small signals far outnumber big ones
  - The relative error of uniform quantization increases with the decrease of signal amplitude

1. Quantitative interval division
2. Quantization level selection

$\{x_i\}_{i=1}^{L+1}$ Hierarchical level

$\{y_i\}_{i=1}^{L}$ Quantization level

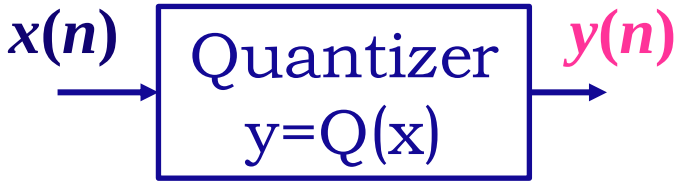
encoder: Hierarchical level — In which range?

decoder: Quantization level — Which level to reconstruct?

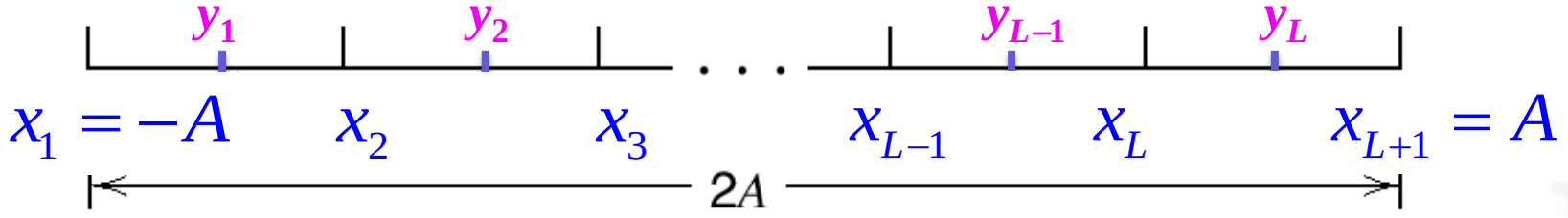
Sample level → k-bit sample encoding → Quantization level

How to optimize the design of encoder and decoder at the same time?

$L = 2^k$



Nonuniform quantization : Nonlinear mapping



Lloyd-Max Quantizer

Iterative Optimization

1. Quantitative interval division
2. Quantization level selection

$\{x_i\}_{i=1}^{L+1}$ Encoder

$\{v_i\}_{i=1}^{L}$ Decoder

1. Given decoder, Optimize encoder nearest neighbor conditions: $I_i=\{y_i \text{ nearest neighbour}\}$

$$x_i = \frac{y_i + y_{i+1}}{2} \qquad x_1 = -\infty \qquad x_{L+1} = \infty \qquad \frac{\partial N_q}{\partial x_i} = 0$$

Find $\{x_i\}_{i=1}^{L+1}$ and $\{y_i\}_{i=1}^{L}$

Minimize $N_q = \sum_{i=1}^{L} \int_{x_i}^{x_{i+1}} (x - y_i)^2 \, p(x) dx$

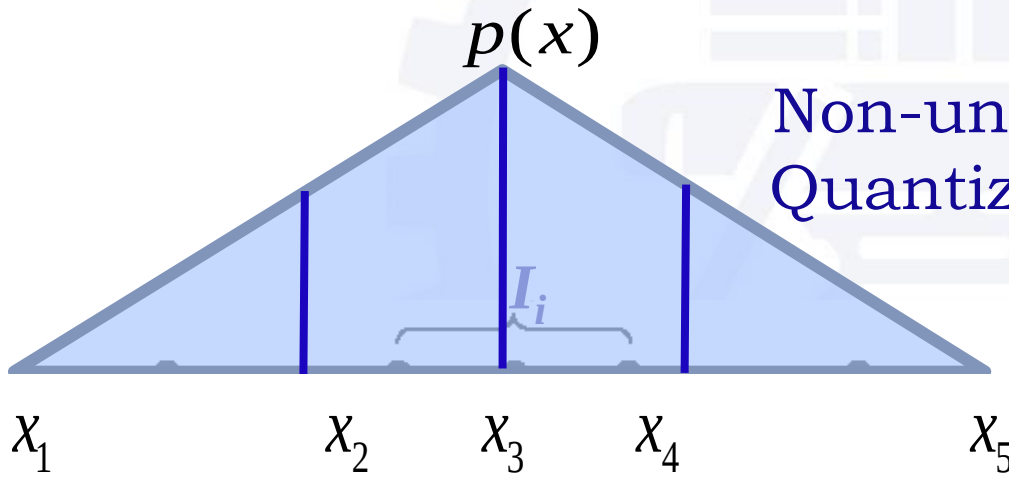
$$(x - y_i)^2 \le (x - y_j)^2, \; j \ne i$$

$\{y_i \text{ nearest neighbour}\}$

2. Given encoder, Optimize the conditional mean requirements of the decoder: y_i=Centroid of I_i

$$y_i = \frac{\int_{x_i}^{x_{i+1}} x p(x) dx}{\int_{x_i}^{x_{i+1}} p(x) dx} \qquad \frac{\partial N_q}{\partial y_i} = 0$$

Steps 1 and 2 alternate until Nq no longer decreases



Non-uniform Quantization

Optimal quantization under uniform distribution

$\{x_i\}_{i=1}^{L+1}$ Encoder $x_i = \dfrac{y_i + y_{i+1}}{2}$

If M is very large, the interval is approximately uniformly distributed and the middle value of the interval can be taken

$\{y_i\}_{i=1}^{L}$ Decoder $y_i = \dfrac{\int_{x_i}^{x_{i+1}} xp(x)dx}{\int_{x_i}^{x_{i+1}} p(x)dx} = \dfrac{x_i + x_{i+1}}{2}$

The optimal quantizer for a uniformly distributed signal is Uniform Quantization:
Equal division division, Median reconstruction

Optimal Quantizer Summary

Quantification process：Divide quantization interval and select quantization level

Iterative Optimization：Quantization level = expected centroid of the interval （E）

Interval belonging = nearest neighbor （M）

When the signal is uniformly distributed, the uniform quantizer is the best quantizer

Coding and decoding of 13 piecewise PCM

PCM performance analysis

4.3 Pulse Code Modulation PCM

4.3.1 A-Law 13 Piecewise Line PCM coding

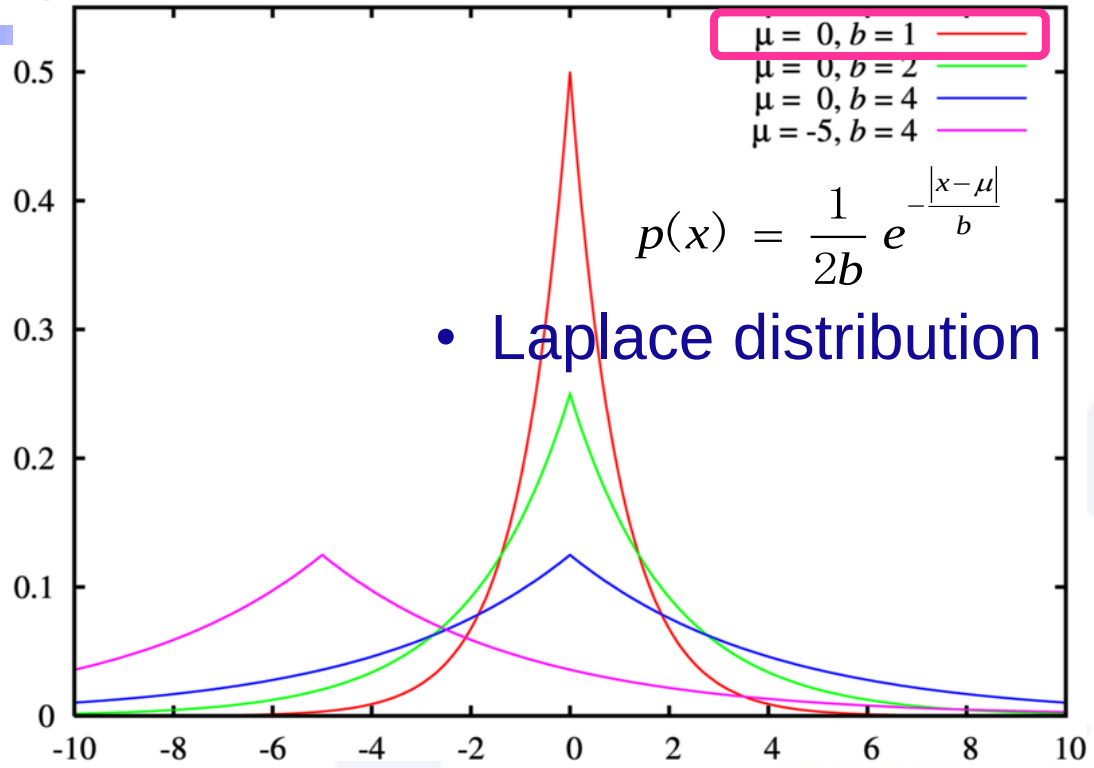
Speech Characteristics and Masking Effects

- Speech signal~Laplace distribution
  - Abundant small signals
  - Large dynamic range

$$p(x) = \frac{1}{\sigma_x \sqrt{2}} e^{-\frac{\sqrt{2}|x|}{\sigma_x}}$$

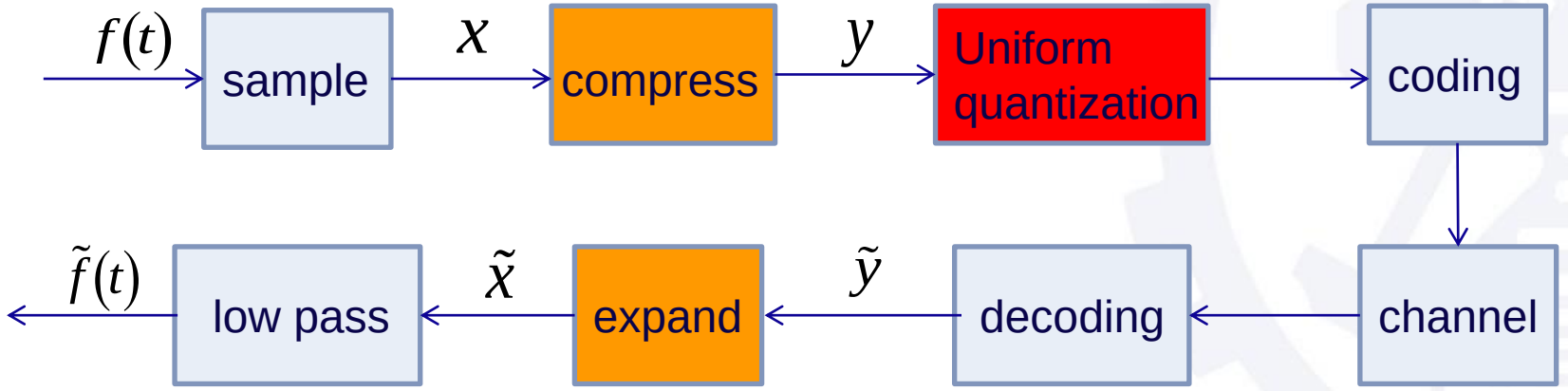
- Ear Masking Effects
  - Sensitive to small signal
  - Insensitive to large signals
- Nonuniform quantization
  - Assign more quantization levels to small signals

= nonlinear mapping +uniform quantization



$$p(x) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}}$$

- Laplace distribution

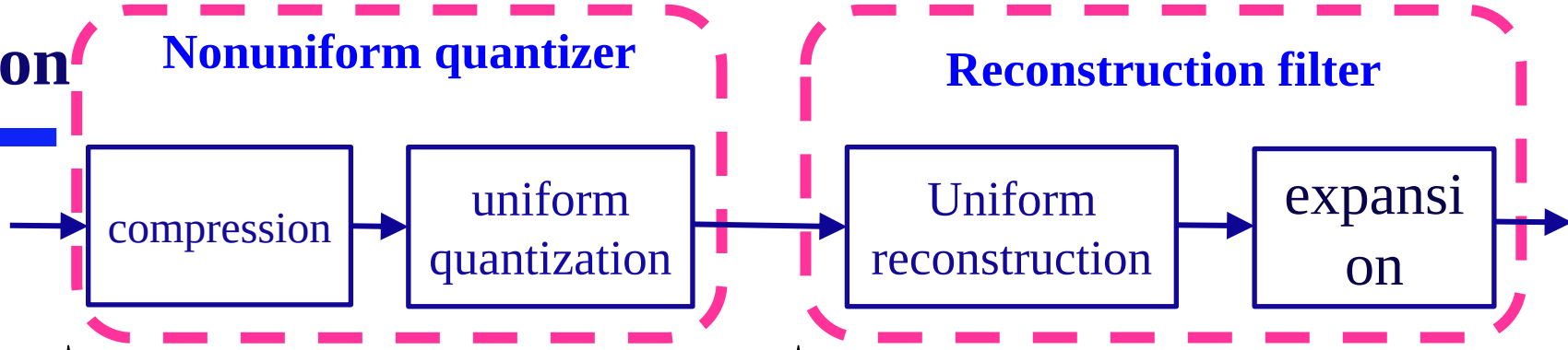


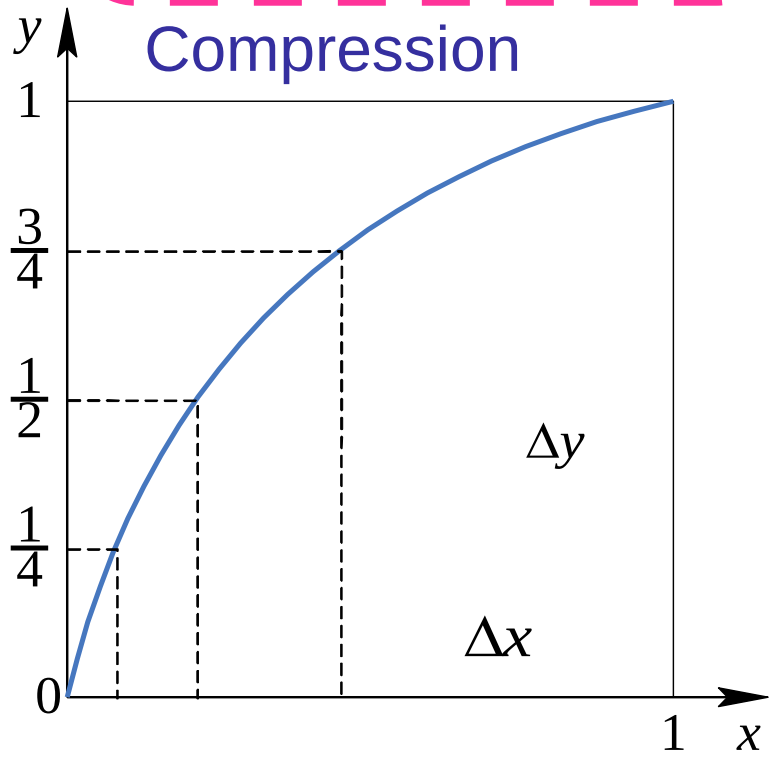
Compression and Expansion

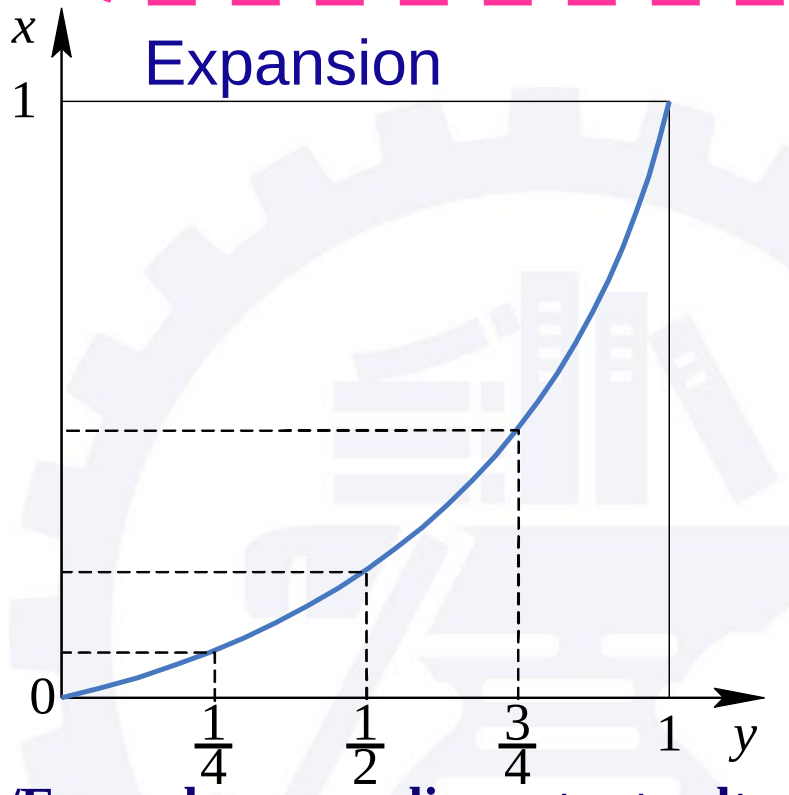
- Nonuniform quantization
  = compression
  + uniform quantization

- Recovery of original signal: Expansion curve

- The Compression Curve and the Expansion Curve are symmetrical with 45 degree bisector







x **— Compressor normalize input voltage/Expander normalize output voltage**

y **— Compressor normalize output voltage/Expander normalize input voltage**

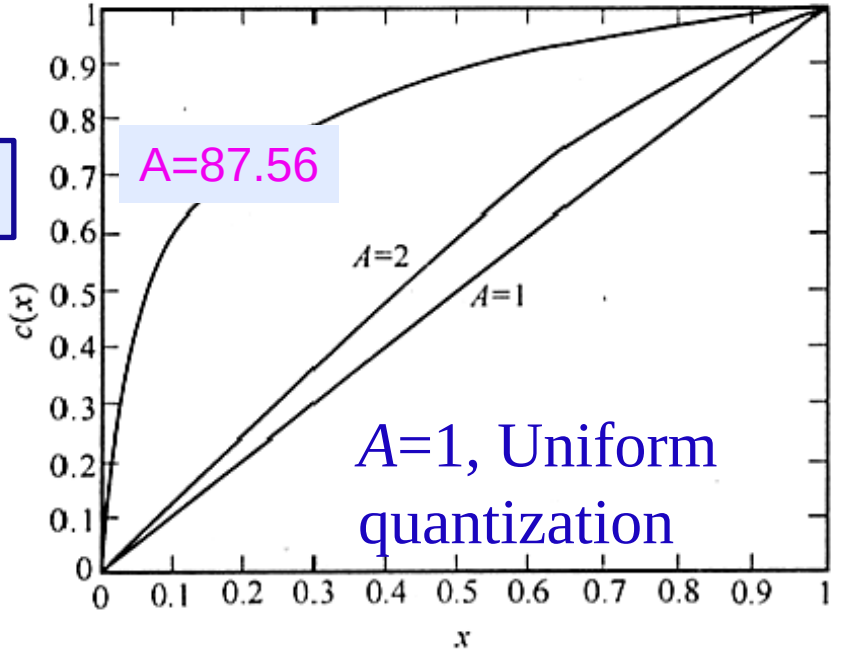
Two logarithmic companding curves

ITU-T G.712

- *A*-Law(Europe and China)： *A*=87.6

$$|y| = \begin{cases} \dfrac{A|x|}{1+\ln A}, & 0 \le x \le \dfrac{1}{A} \\ \dfrac{1+\ln(A|x|)}{1+\ln A}, & \dfrac{1}{A} \le x \le 1 \end{cases}$$

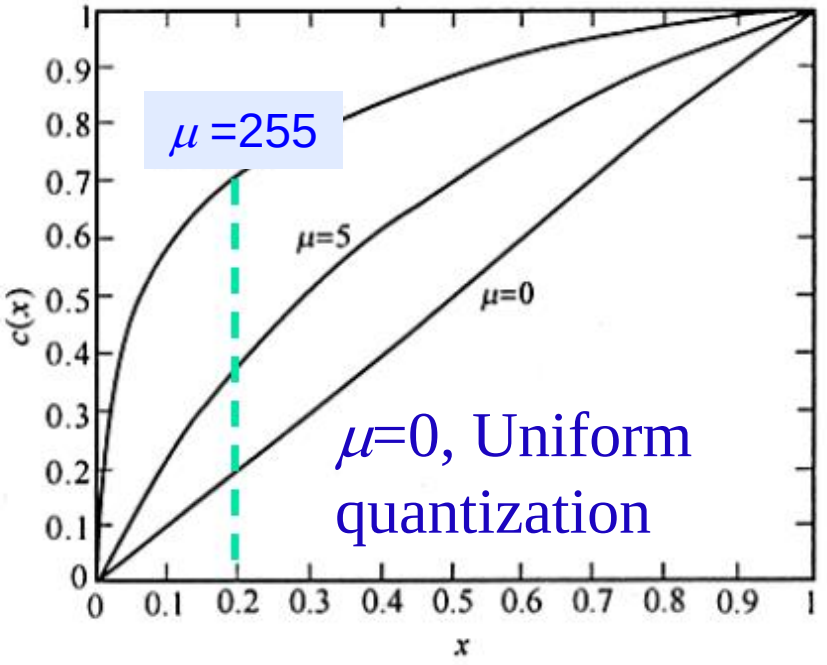
When small signals, the signal-to-noise ratio increases 24dB



- μ-*L*aw(North America and Japan)： μ =255

$$|y| = \frac{\ln(1+\mu|x|)}{\ln(1+\mu)}$$

When small signals, the signal-to-noise ratio increases 33.5dB



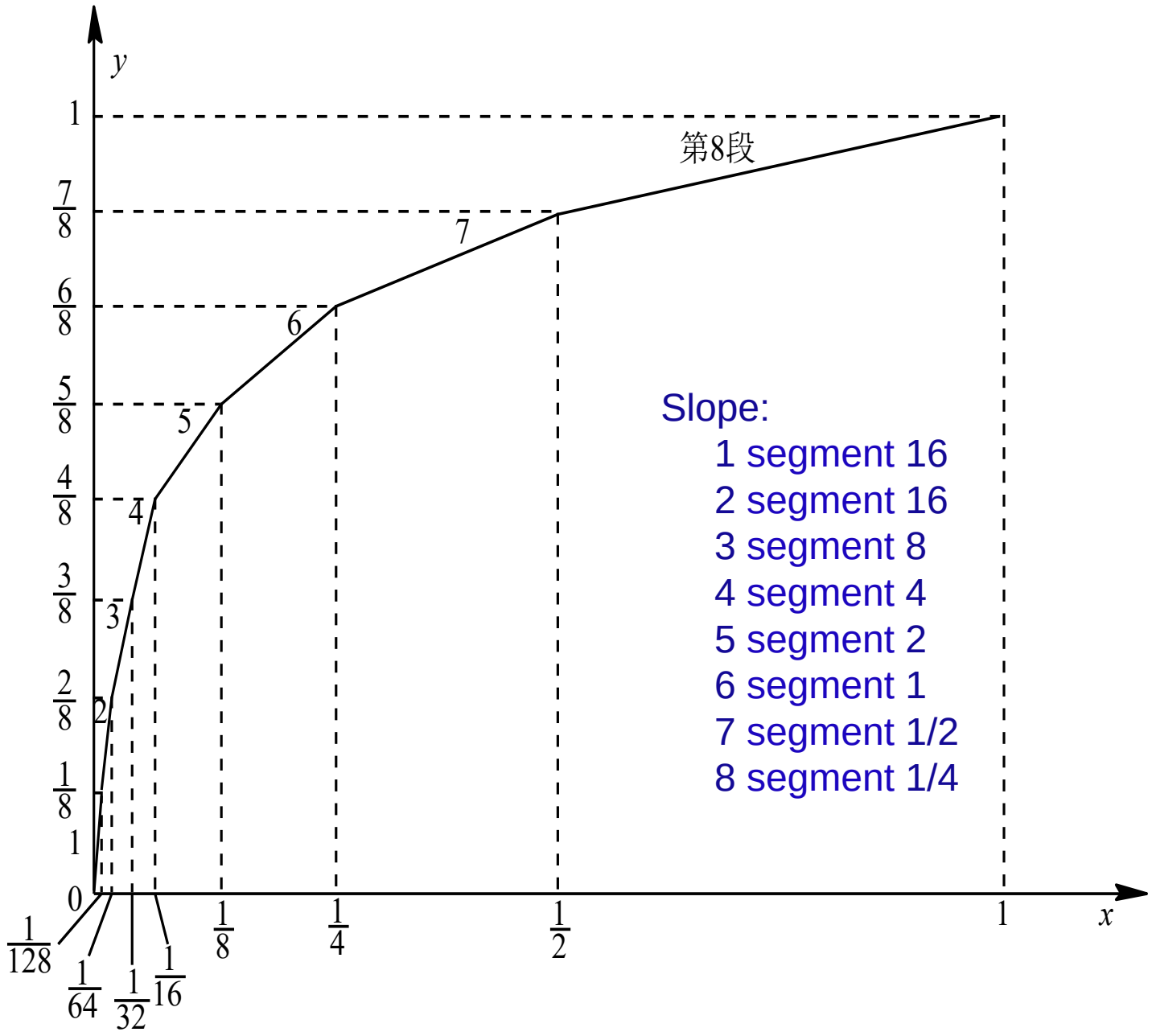
Compare the performance of A-Law and μ - Law in large and small signal range

The larger the values of constants A and μ, the stronger the degree of companding

A-Law 13 piecewise approximation

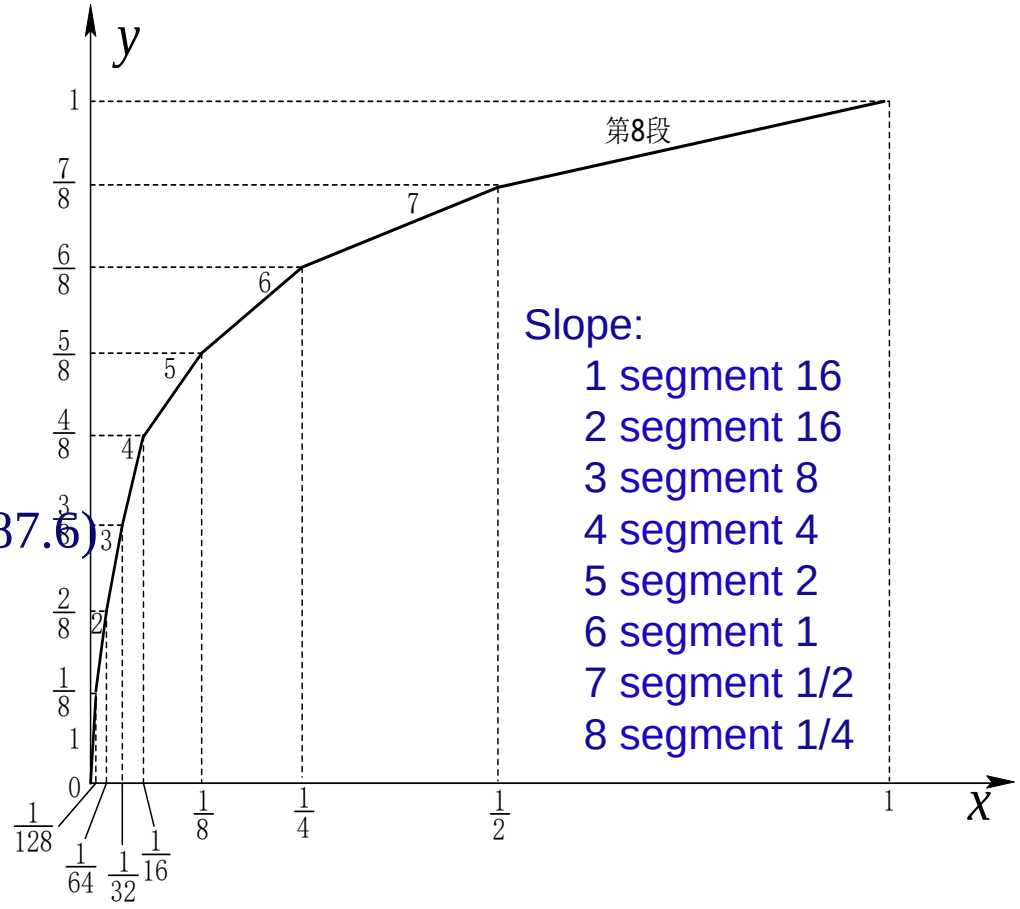
$$y=\begin{cases}\dfrac{Ax}{1+\ln A}, 0\le x\le \dfrac{1}{A}\\ \dfrac{1+\ln(Ax)}{1+\ln A}, \dfrac{1}{A}<x\le 1\end{cases}$$

- Problem: continuous smooth curve, real-time calculation is complex
- Finding approximate realization
- Nonuniform segmentation: 8 segments
- *x*: Bisection division
- *y*: Uniform division



A-Law 13 piecewise approximation

- 13 piecewise
  - 16 sections in the positive and negative range
  - 13 different slopes
- 13 piecewise line and A-Law compression characteristics (A = 87.6)
  - Only compare at each turning point and end point of the lines
  - The approximation error is very small



- 15 piecewise approximation of μ-Law characteristics

Line segment number *i*	1	2	3	4	5	6	7	8
Starting level of line segment y_{i-1}	0	1/8	2/8	3/8	4/8	5/8	6/8	7/8
A-Law line segment level x_{i-1}	0	1/128	1/60.6	1/30.6	1/15.4	1/7.79	1/3.93	1/1.98
13 line segment layering level x_{i-1}	0	1/128	1/64	1/32	1/16	1/8	1/4	1/2
Slope of lines	16	16	8	4	2	1	1/2	1/4

Coding scheme

- The rule of mapping discrete sample value into a codeword after sampling quantization of analog signal

- folded binary code, FBC
  - The highest bit represents the voltage polarity
  - The other bits represent absolute amplitude
  - The absolute amplitude is mapped naturally
  - Bit error has little effect on small signal

Decimal	Natural code	Folding code	Gray code
15	1111	1111	1000
14	1110	1110	1001
13	1101	1101	1011
12	1100	1100	1010
11	1101	1011	1110
10	1010	1010	1111
9	1001	1001	1101
8	1000	1000	1100
7	0111	0000	0100
6	0110	0001	0101
5	0101	0010	0111
4	0100	0011	0110
3	0011	0100	0010
2	0010	0101	0011
1	0001	0110	0001
0	0000	0111	0000

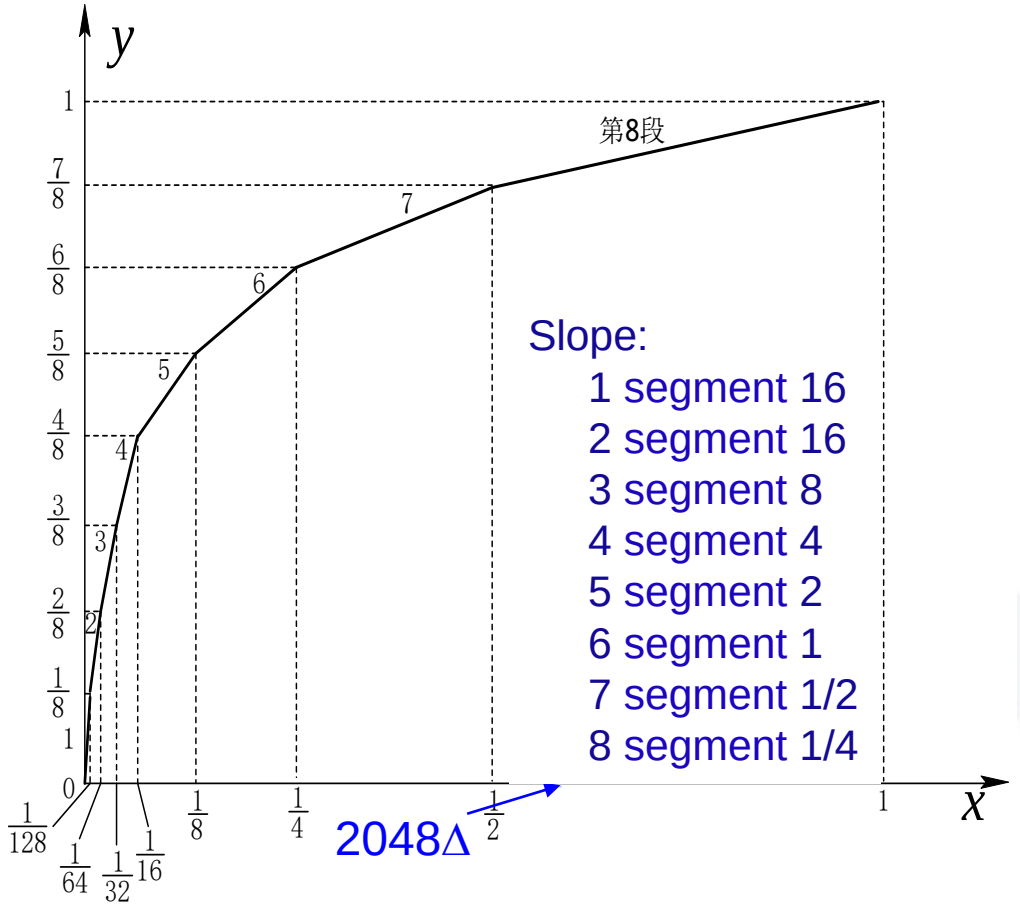
Encoding and decoding

Polarity code　Paragraph code　Segment code

C_7　　$C_6C_5C_4$　　$C_3C_2C_1C_0$

positive： 1
negative： 0

Equally divided into 16 segments in each paragraph



- Coding implementation

  Successive comparison: the thought of "half comparison"

  Start level coding: the coding process only needs to determine the section

- Decoding implementation

  Quantization level: direct decoding result plus $\Delta_i/2$ to reduce quantization noise

Paragraph number i	1	2	3	4	5	6	7	8
Paragraph code	000	001	010	011	100	101	110	111
Starting level $I_i(\Delta)$	0	16	32	64	128	256	512	1024
segment quantization interval $\Delta_i(\Delta)$	1	1	2	4	8	16	32	64

Example question

The sampling value of input signal is 1270 quantization units，1270Δ，According to the characteristics of A-Law 13 line coding, calculate quantization error.

Answer：

（1）Determine polarity code. Since the sampling value is positive, M_1=1。

（2） Determine paragraph code. Since the sampling value of input signal 1270Δ>1024Δ，$M_2M_3M_4$=111

$$= \lfloor (1270\Delta - 1024\Delta)/\Delta_k \rfloor$$
$$= \lfloor 246\Delta/64\Delta \rfloor = 3$$

（3）Determine segment code。Number of levels in the paragraph

Segment code is 0011，

Codeword：11110011

（4）Quantization error

$q = m - v$，m = 1270 Δ

v = 1024Δ+3.5*64 Δ

q = 22Δ

Paragraph number i	1	2	3	4	5	6	7	8
Paragraph code	000	001	010	011	100	101	110	111
Starting level $I_i(\Delta)$	0	16	32	64	128	256	512	1024
segment quantization interval $\Delta_i(\Delta)$	1	1	2	4	8	16	32	64

Summary of A-Law 13 piecewise line coding

13 line approximation： x division, y Uniform division

Folded coding scheme is adopted

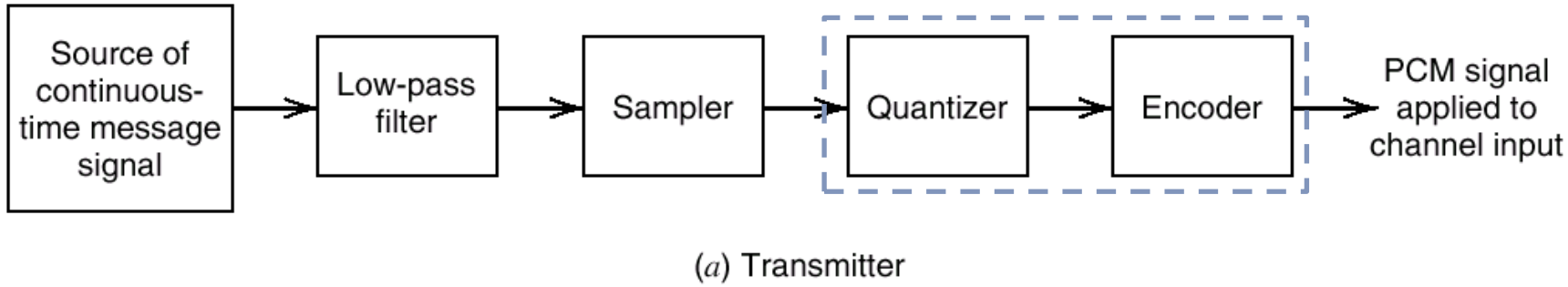
Start level coding, median level decoding

4.3.2 PCM Performance Analysis

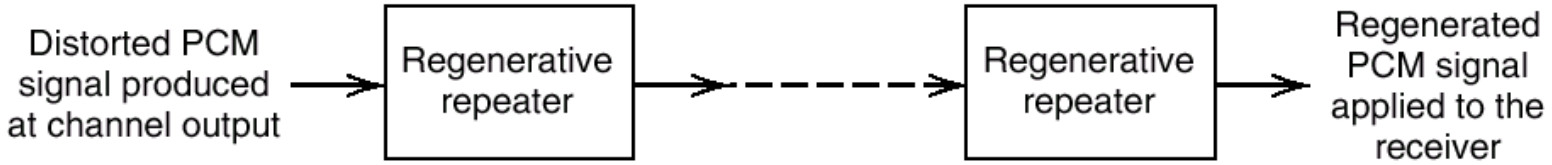
Pulse code Modulation System

Pulse code Modulation

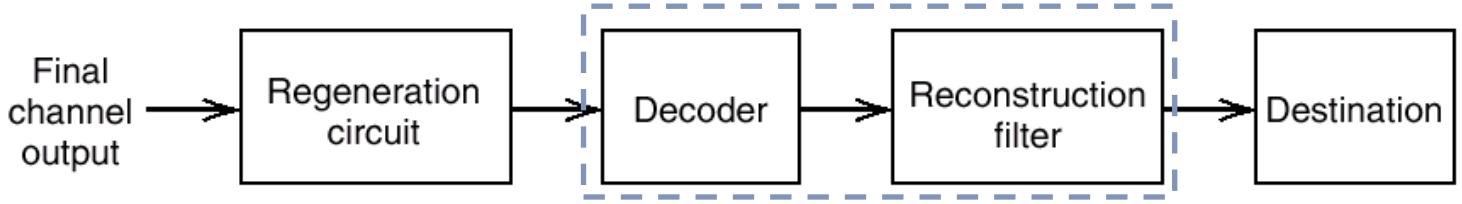
- Actual compression coding method of speech signal



(*a*) Transmitter



(*b*) Transmission path



(*c*) Receiver

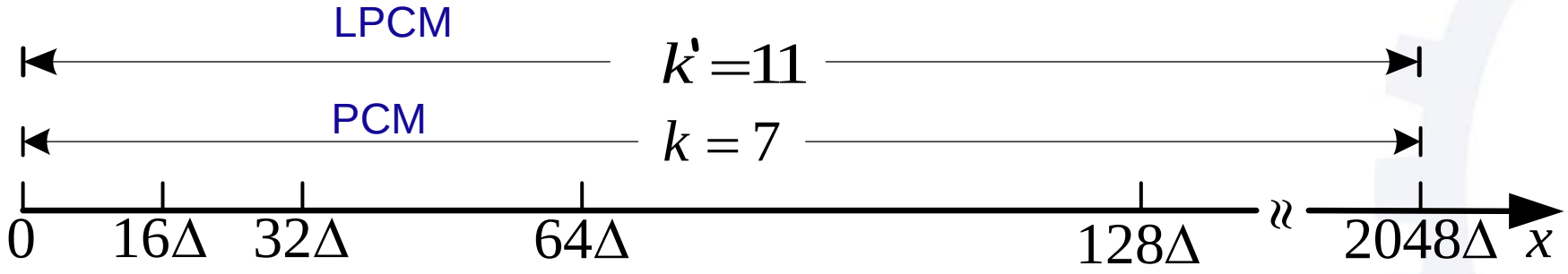
- 2 main steps:
  - Sampling： f_s=8000Hz
  - Quantization & coding：
    - non-uniform quantization, each sample value is quantized to 8 bits

- PCM coding rate 64kbps

Efficiency: number of encoded bits

- Take the minimum quantization interval in 13 broken lines as the uniform quantization interval
  - Dynamic range：（–2048 Δ, 2048 Δ)
  - Ensure that the quantization interval Δ is equal——the signal-to-noise ratio of small signal is the same
    - The uniform quantization need 12bit
    - The non-uniform quantization only need 8bit

(LPCM)linear PCM：
uniform quantization
(Logarithmic companding) PCM：
non-uniform quantization



number of coding bits required for amplitude part after excluding polarity bits

Analysis of anti noise performance

- Noise sources
  - Quantization noise(N_q)
  - Bit error noise(N_e)

1)AWGN
2)Only 1 bit error
3) uniform distribution of signal amplitude

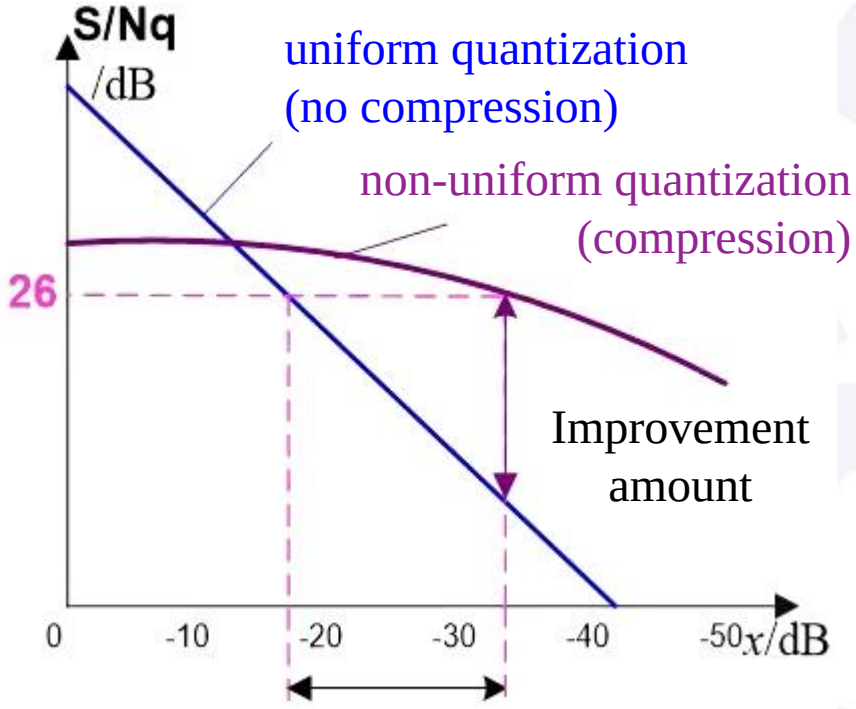
- System total SNR: $\frac{S}{N} = \frac{S}{N_q + N_e} = \frac{L^2}{1 + 4L^2 P_e}$

Small SNR: $\frac{S}{N} \approx \frac{1}{4P_e}$ Bit error distortion is the main influence

Large SNR: $\frac{S}{N} \approx L^2$ Quantization noise is the main influence

For example: $\left.\begin{matrix} 4L^2 P_e = 1 \\ L = 256 \end{matrix}\right\} \Rightarrow P_e \approx 3.8 \times 10^{-6}$

- Quantized signal to noise ratio
  - uniform quantization
  - non-uniform quantization



Increment of dynamic range

Limitation and improvement of PCM

- Limitation
  - High complexity
  - Compared with analog signal transmission, the system bandwidth increases
- Improvement
  - Reduce complexity
    - Delta modulation ΔM, the cost is to increase the system bandwidth
  - Predictive coding reduces bandwidth, the cost is to increase the complexity
    - Differential pulse code modulation DPCM
    - Self-adaption differential pulse code modulation ADPCM

PCM Performance Conclusion

PCM noise comes from quantization process and channel transmission.

PCM coding is more efficient than LPCM, but bandwidth is significantly increased compared with analog transmission.

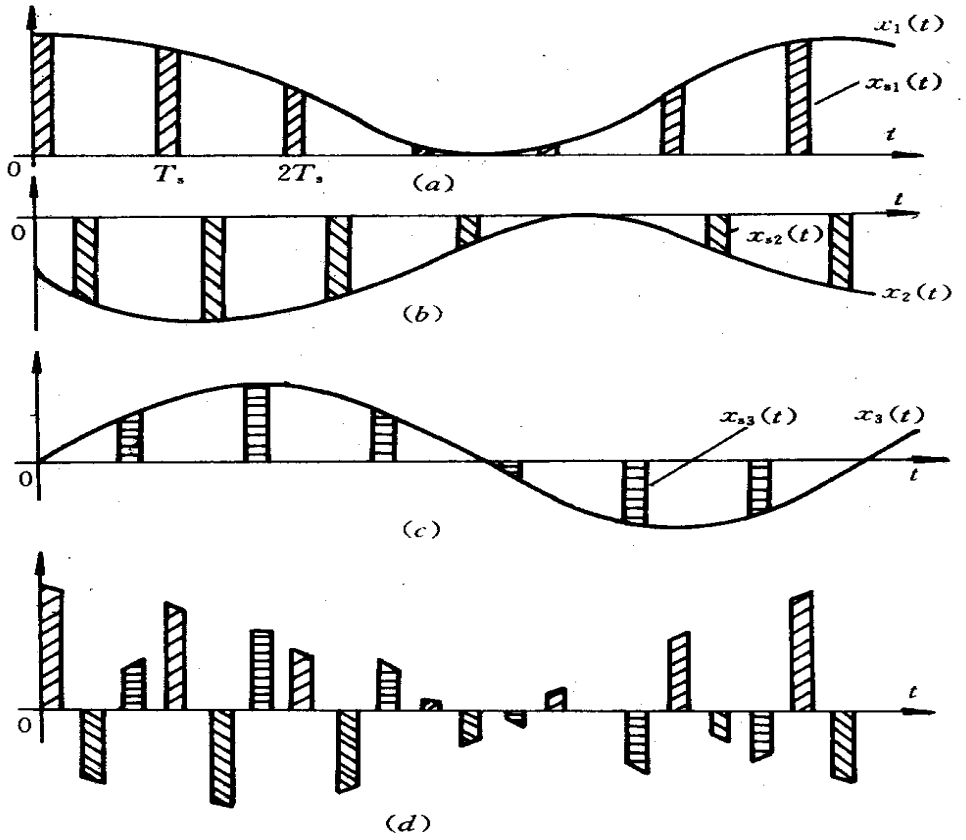
4.3.3 Time Division Multiplexing

Time Division Multiplexing(TDM)

Time division multiplexing

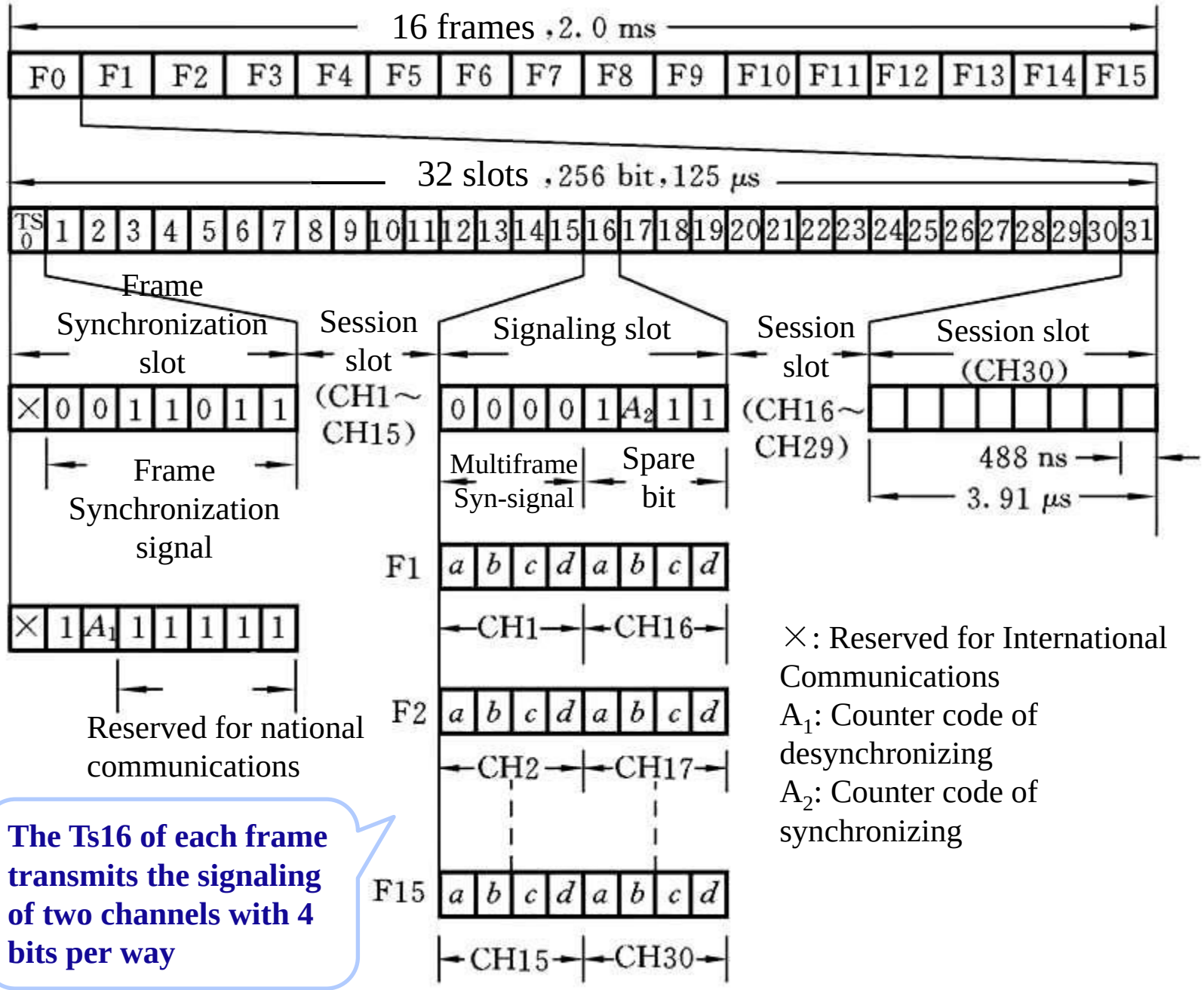
- The transmission time is divided into several non-overlapping time slots, and each signal occupies its own time slot in a certain order.

- Advantage of TDM
  - Suitable for digital transmission
  - Reduction of nonlinear requirements
  - Strong anti-interference ability



Frame structure of Basic group

- E-Series basis group E1
- 30/32 ways PCM
- Slot: Minimum time unit
- Frame: 1 sampling interval
- Multiframe: 16 frames
  - The signaling information of each session is transmitted once per complex frame.
- Basic group signal rate：2048kb/s



×: Reserved for International Communications

A_1: Counter code of desynchronizing

A_2: Counter code of synchronizing

The Ts16 of each frame transmits the signaling of two channels with 4 bits per way

T1 (North America) Frame structure ?

Rate calculation of multiplex signal?

Bits per frame

TDM Conclusion

By introducing the concept of time slot and using frame structure design, the physical channel can be reused effectively.

The multiplexing signal contains service bits and control bits.

DPCM and ADPCM

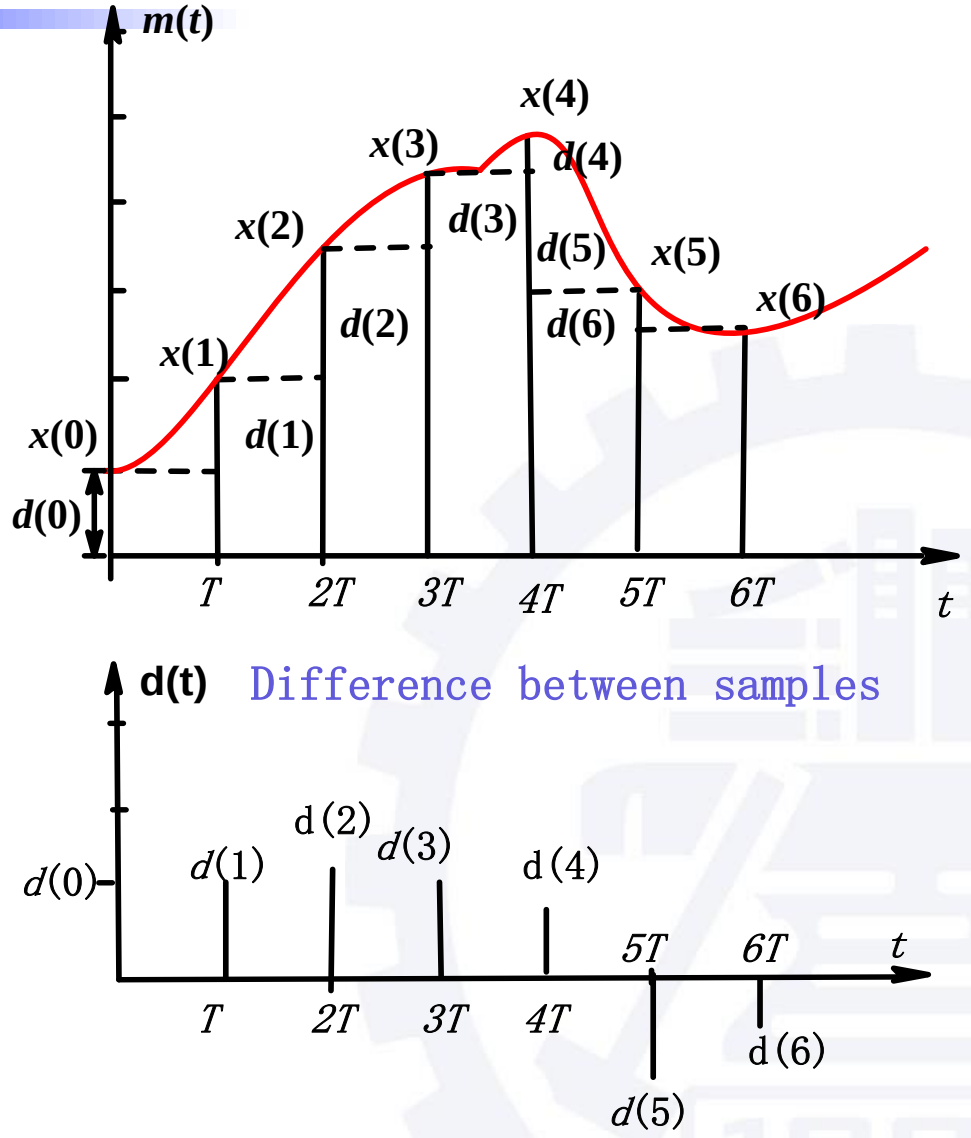
Delta modulation ΔM

4.4 Predictive coding

4.4.1 DPCM and ADPCM

Differential Pulse Code Modulation（DPCM）

- The fundamental reason for the inefficiency of PCM is that each sample value is encoded independently, while the whole amplitude range of the sample value requires more bits of coding.
- There is correlation between adjacent sample values. Predict the next sample value based on history——prediction coding
- The basic idea of DPCM： to encode the prediction error, which is the difference between adjacent samples.
- It can reduce the number of coding bits and reduce redundancy.



Realization of DPCM

- Coding

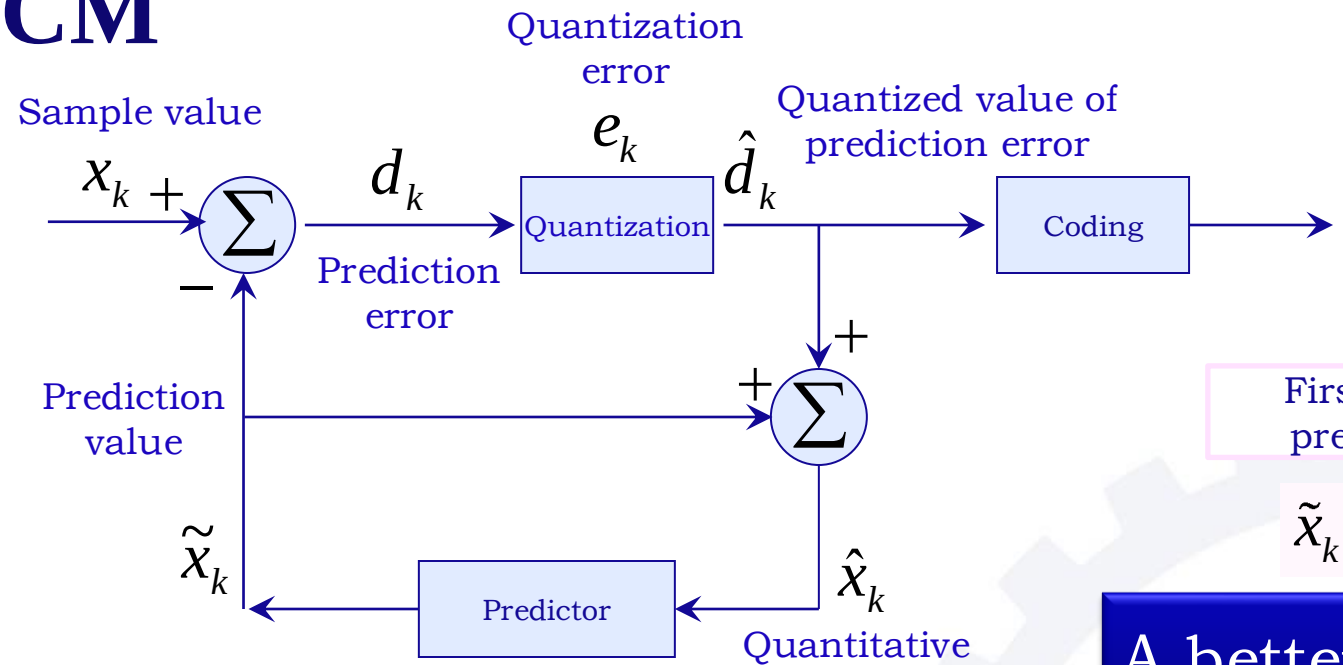
$$d_k = x_k - \tilde{x}_k$$

$$\hat{d}_k = d_k - e_k$$

$$\hat{x}_k = \tilde{x}_k + \hat{d}_k$$

$$\hat{x}_k = x_k - e_k$$

Sample quantitative error = difference quantization error

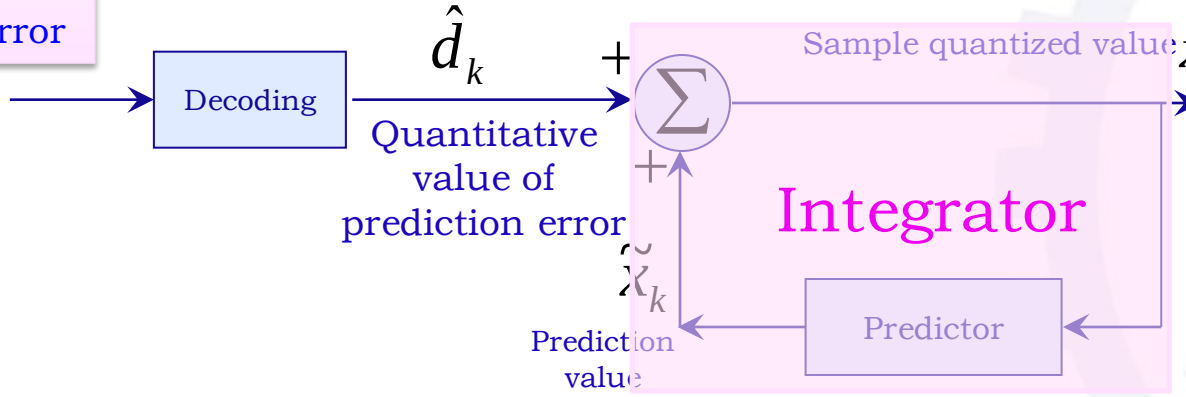


First-order prediction

$$\tilde{x}_k = \hat{x}_{k-1}$$

A better predictor?

- Decoding



$$\hat{x}_k = \tilde{x}_k + \hat{d}_k = \hat{x}_{k-1} + \hat{d}_k$$

$$= \sum_{i=0}^{k} \hat{d}_i$$

Adaptive Differential Pulse Code Modulation (ADPCM)

- First-order Prediction $\tilde{x}_k = \hat{x}_{k-1}$
- N-order Prediction $\tilde{x}_k = \sum_{i=1}^{N} W_i \hat{x}_{k-i}$

The best linear prediction under MMSE criterion

- ADPCM
  - The speech signal is essentially non-stationary.
  - Adjust the weight coefficient adaptively according to the signal in real time
  - Adjust the Quantization accuracy adaptively according to the signal in real time
- Quantized signal-to-noise ratio

$$S/N = \frac{E[x_k^2]}{E[e_k^2]} = \frac{E[x_k^2]}{E[d_k^2]} \cdot \frac{E[d_k^2]}{E[e_k^2]} = G_p \cdot \left(\frac{S}{N}\right)_q$$

Bit Rate : 32 kbits/s (4bits/sample)

G_p Prediction Gain

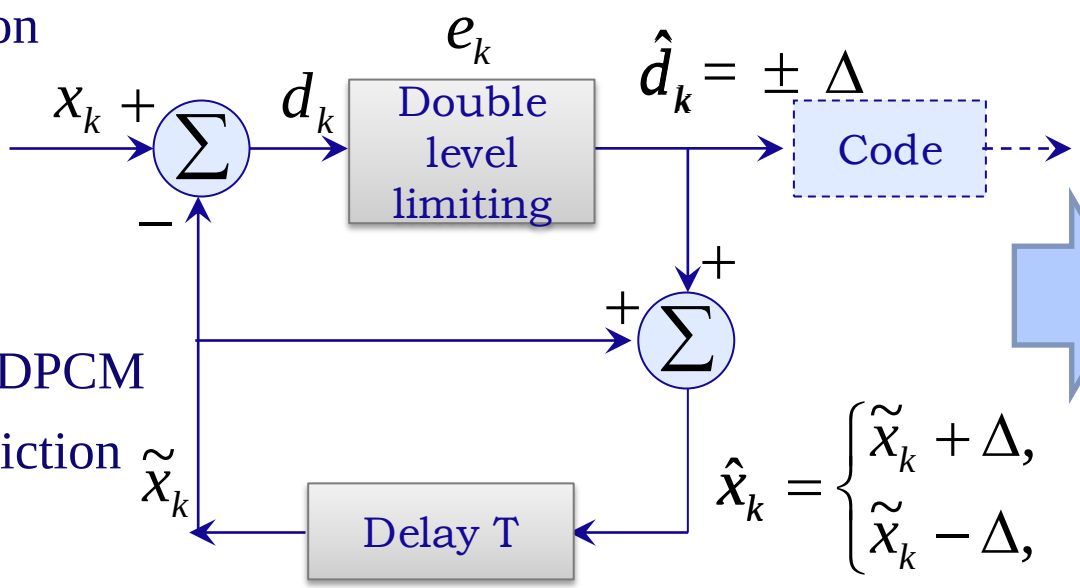
DPCM and ADPCM Summary

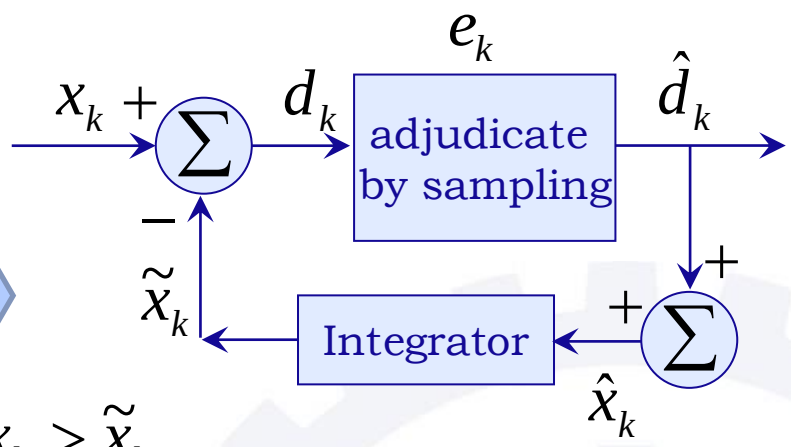
- DPCM encodes the prediction error by taking the adjacent sample values as the prediction value, and the prediction gain can be obtained.
- ADPCM adaptively adjusts the weight coefficient of the predictor or the quantization interval of the quantizer to improve the prediction gain.

4.4.2 Delta Modulation

Delta Modulation

- Delta Modulation
  (DM，ΔM)

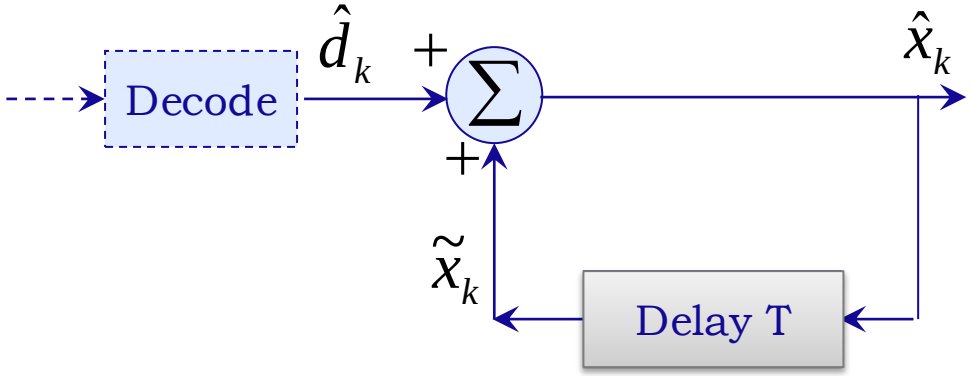


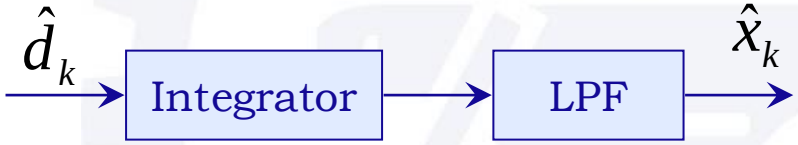


- Special case of DPCM
- First-order Prediction
- 1-bit Coding

$$\hat{x}_k = \begin{cases} \tilde{x}_k + \Delta, & x_k > \tilde{x}_k \\ \tilde{x}_k - \Delta, & x_k < \tilde{x}_k \end{cases}$$

Codec can be omitted



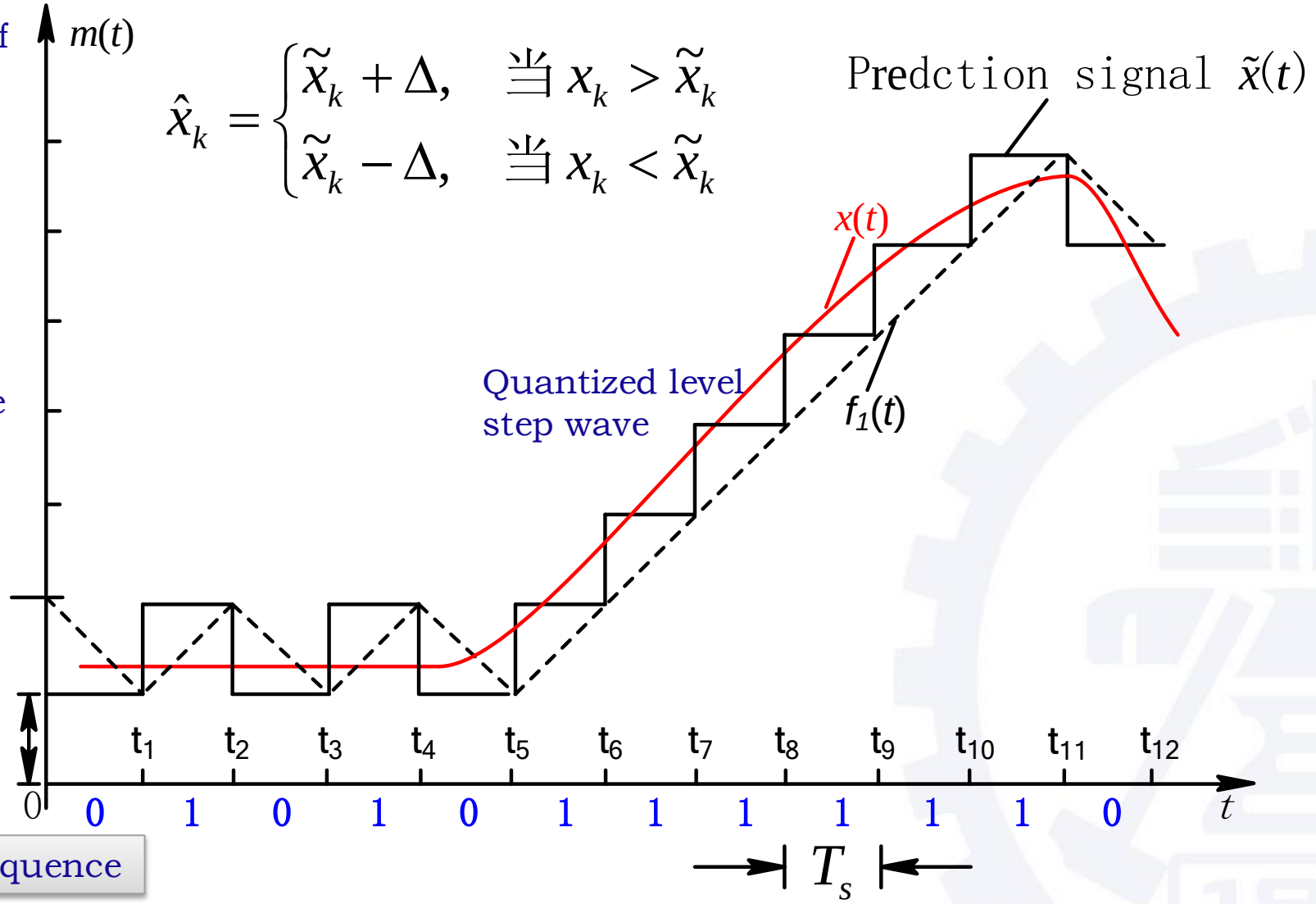


Coding Process

- Encode the change of the sample value by 1-bit coding

Advantages:
- Simple encoding
- Simple decoding

Disadvantages:
- Small dynamic range



Coding sequence

Quantization Noise

$$\frac{\Delta}{T_s} \geq \left|m'(t)\right|_{\max}$$ The sampling rate f_s is much larger than the Nyquist rate

- **Slope overload distortion: the signal changes too fast, the step wave can not keep up with the change.**
- **Overload noise**
- **Particle noise**
- **Slope non-overload condition**
  - Minimum sampling rate
  - Maximum allowable coding level (critical overload voltage)
- **Prevent no load**
  - Minimum coding level
- **Coding dynamic range**

Adaptive adjustment of quantization accuracy

$$f_s \geq \frac{\left|m'(t)\right|_{\max}}{\Delta} = 2f_m \frac{\pi A}{\Delta}$$

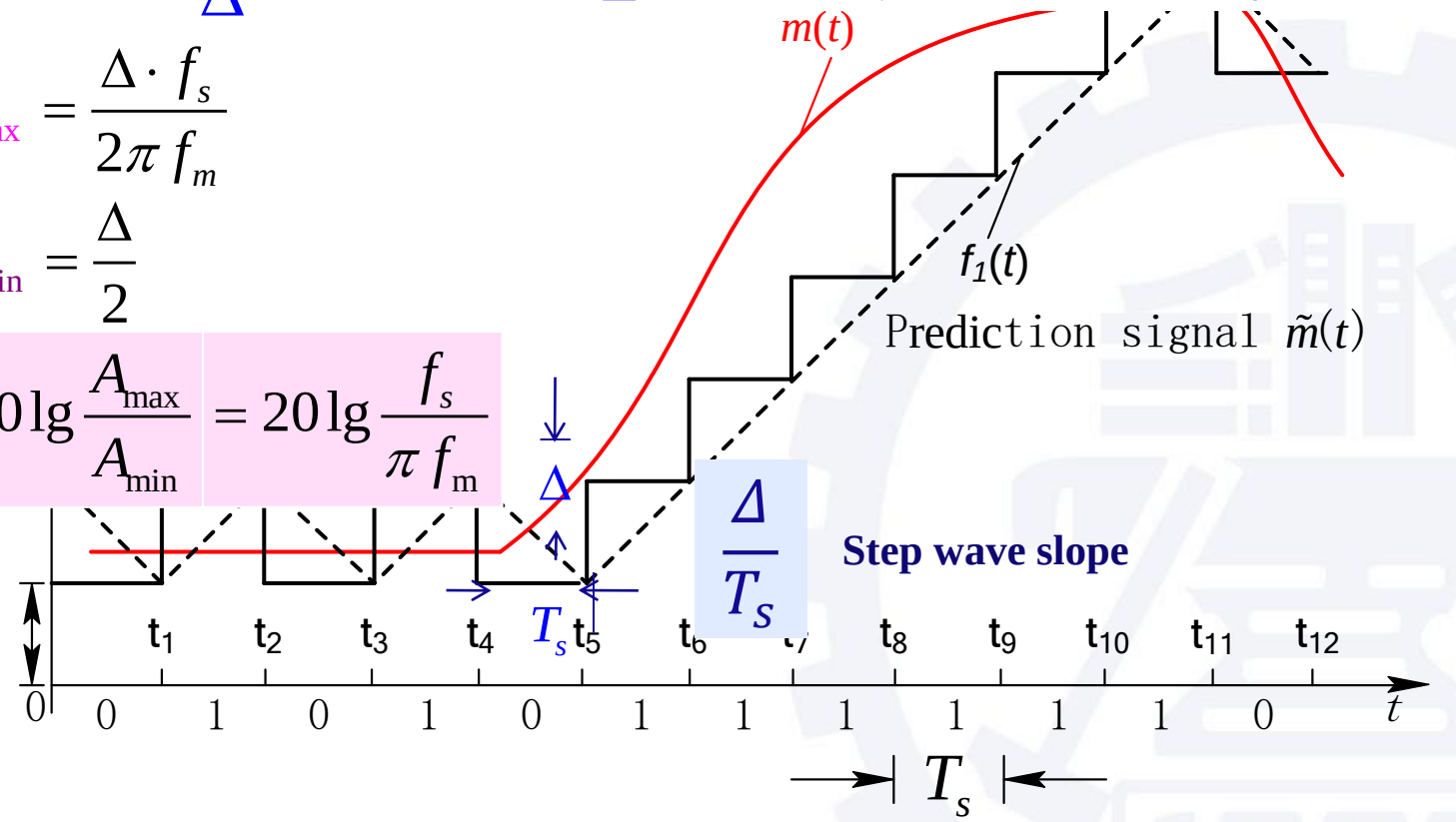
$$m(t) = A\cos\omega_m t$$

$$\left|m'(t)\right|_{\max} = 2\pi A f_m$$

$$A_{\max} = \frac{\Delta \cdot f_s}{2\pi f_m}$$

$$A_{\min} = \frac{\Delta}{2}$$

$$D_c = 20\lg\frac{A_{\max}}{A_{\min}} = 20\lg\frac{f_s}{\pi f_m}$$



Quantized Signal-to-Noise Ratio

Quantization noise $e_k = \hat{d}_k - d_k \in [-\Delta, \Delta]$ $N_q = \frac{\Delta^2}{3}$ $n_0 = \frac{\Delta^2}{3 f_s}$

Particle noise

It is approximately considered that the quantization noise is uniformly distributed at (0, f_s).

Noise power via LPF $N_o = n_0 B = \frac{\Delta^2}{3 f_s} f_H$

f_H: LPF Cutoff frequency

Signal $m(t) = A\cos\omega_m t$

$A_{\max} = \frac{\Delta \cdot f_s}{2\pi f_m}$

Signal Power $S_o = \frac{A^2}{2} = \frac{\Delta^2 f_s^2}{8\pi^2 f_m^2}$

Quantized Signal-to-Noise Ratio

$$\frac{S_o}{N_o} = \frac{3}{8\pi^2} \cdot \frac{f_s^3}{f_m^2 f_H} \approx 0.04 n^3$$

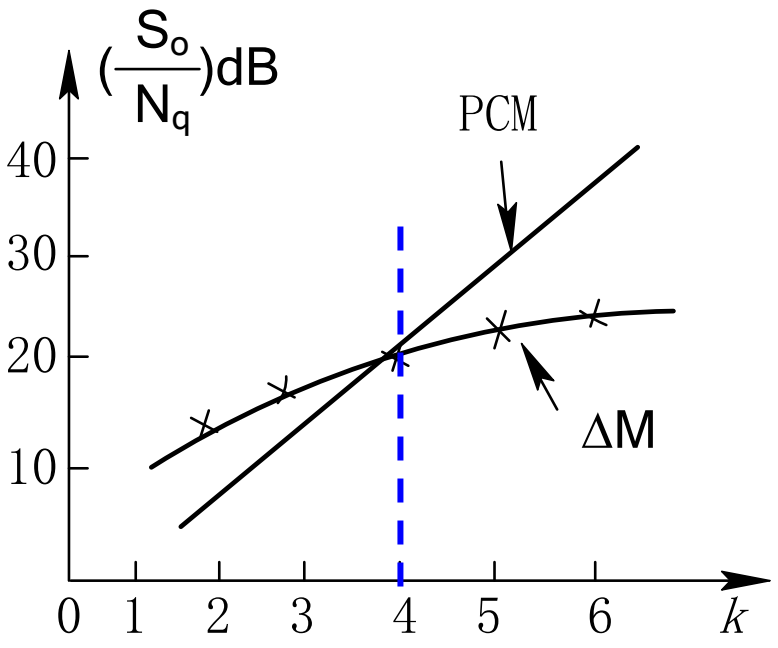
$$\left(\frac{S_o}{N_o}\right)_{\text{dB}} = 30\lg f_s - 20\lg f_m - 10\lg f_H - 14 \approx 30\lg n - 14$$

$f_H = f_m$

$f_s = n f_m$

- There is a square relationship between quantized signal-to-noise ratio and sampling frequency f_s. When f_s increases by one time, quantized signal-to-noise ratio increases 9dB.
- n is large enough to meet the general communication quality requirements

Comparison between PCM and ΔM



Number of coded bits

	PCM	ΔM
Number of coded bits	k, 8	1
f_s sample rate kHz (R_b fixed)	$2f_m$, 8	$f_s' \geq \frac{\lvert m'(t)\rvert_{\max}}{\Delta} \sim kf_m$
Bit Rate	kf_s	f_s'
Main Lobe Bandwidth kHz (Same quantity criterion)	64	100
Quantization dB (R_b fixed)	$6k$	$10\lg(0.32k^2\frac{f_H^2}{f_m^2})$
Channel error influence	Sensitive, serious in high bit 10^{-5}~10^{-6}	Insensitive 10^{-3}~10^{-4}
Equipment complexity	Complex large capacity trunk communication	Simple, small capacity and low quality military communication

Delta Modulation Summary

- 1-bit predictive coding
- There are particle noise and overload noise.
- The sampling is much higher than that of PCM, in order to ensure that the slope is not overloaded.

Quiz

1. In order to realize digital transmission of analog signals, which 3 steps are required to digitize the analog signal?

2. What is the sampling theorem for a low-pass signal?

3. How to avoid the aliasing phenomenon while sampling the analog signal?

4. What is the distortion-free sampling rate for a band-pass signal?